

4

NUMBER AND ALGEBRA

PRODUCTS, FACTORS AND SURDS

In 825 CE, the Persian mathematician al-Khwarizmi used the Arabic word ‘al-jabr’ to describe the process of adding equal quantities to both sides of an equation. When al-Khwarizmi’s book was translated into Latin and introduced to Europe, ‘al-jabr’ became ‘algebra’ and the word was adopted as the name for the branch of mathematics that uses formulas to describe patterns and relationships in our world. The photo above shows a monument of al-Khwarizmi located in Khiva, Uzbekistan.



Chapter outline

		Working mathematically				
4.01	The index laws	U	F		R	C
4.02	Fractional indices*	U	F		R	C
4.03	Adding and subtracting algebraic fractions	U	F			
4.04	Multiplying and dividing algebraic fractions	U	F			
4.05	Expanding and factorising expressions	U	F			
4.06	Expanding binomial products	U	F		R	C
4.07	Factorising special binomial products*	U	F			
4.08	Factorising quadratic expressions $x^2 + bx + c$*	U	F		R	
4.09	Factorising quadratic expressions $ax^2 + bx + c$*	U	F		R	
4.10	Mixed factorisations*	U	F		R	
4.11	Factorising algebraic fractions*	U	F		R	
4.12	Simplifying surds*	U	F		R	
4.13	Adding and subtracting surds*	U	F			
4.14	Multiplying and dividing surds*	U	F			
4.15	Binomial products involving surds*	U	F			
4.16	Rationalising the denominator*	U	F			

***PATH**

Wordbank

binomial product An algebraic expression showing 2 or more binomials multiplied together, for example, $(x + 9)(3x - 4)$

index laws Rules for simplifying algebraic expressions involving powers of the same base, for example, $a^m \div a^n = a^{m-n}$

perfect square A square number or an algebraic expression that represents one, for example, 64, $(x + 9)^2$

quadratic trinomial An algebraic expression that consists of 3 terms, for example, $x^2 + 2x + 6$

rationalise the denominator To simplify a fraction involving a surd by making its denominator rational (that is, not a surd)

simplify a surd To write a surd $\sqrt{x}$ in its simplest form so that x has no factors that are perfect squares

surd A square root (or other root) that is irrational so its exact decimal value cannot be found

Quiz
Wordbank 4

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Videos (20):

4.01 Simplifying with the index laws • Index laws • Numbers and powers • Negative indices
4.02 Fractional indices
4.03 Adding and subtracting algebraic fractions • Algebraic fractions
4.04 Algebraic fractions
4.05 Factorising expressions 2
4.06 Expanding binomial products 1 • Expanding binomial products 2 • Special binomial products • Difference of 2 squares
4.07 Difference of 2 squares
4.08 Factorising quadratic expressions 1
4.09 Factorising quadratic expressions 2 • Factorising quadratic expressions (Advanced) • Advanced algebra
4.12 Simplifying surds 2 • Simplifying surds 1
4.15 Binomial products involving surds
4.16 Rationalising the denominator

Quizzes (5):

- Wordbank 4
- SkillCheck 4
- Mental skills 4
- Language of maths 4
- Test yourself 4

Skillsheets (3):

4.05 Algebra using diagrams • HCF by factor trees • Factorising using diagrams

Worksheets (13):

4.01 Index laws review
4.04 Algebraic fractions
4.05 Expanding and factorising • Algebra 4
4.06 Binomial products • Area diagrams • Mixed expansions • Algebra 2 • Algebra 6 • Special products
4.11 Simplifying algebraic fractions
4.14 Multiplying and dividing surds
Language of maths Algebra 6
Mind map Products, factors and surds

Puzzles (24):

4.01 Indices puzzle • Indices squaresaw
4.02 Indices squaresaw (Advanced) • Exponential equations
4.03 Algebraic fractions puzzle
4.04 Algebraic fractions • Upside-down fractions
4.05 Factorising puzzle • The distributive law
4.06 Expanding brackets • Trinominoes • Products and factors squaresaw • Factorominoes • Enough time
4.07 Grouping • Difference of 2 perfect squares
4.09 Products and factors squaresaw • Perfect squares • Factorominoes • Trinominoes • Factorising puzzle (Advanced)
4.10 Factorominoes • Mixed factorisations
4.12 Simplifying surds
4.13 Surds code puzzle
4.14 Surds
Language of maths Algebra crossword • Surds crossword

Technology (3):

4.06 Expanding binomials
4.09 Factorising trinomials
4.12 Simplifying surds quiz

Spreadsheets (3):

4.04 Expanding binomials
4.09 Factorising trinomials
4.12 Simplifying surds quiz

Presentation (1):

4.01 Index laws

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- (PATH) apply index laws to algebraic expressions with integer indices
- (PATH) apply index laws to algebraic expressions with fractional indices
- add, subtract, multiply and divide simple algebraic fractions
- (PATH) add and subtract algebraic fractions with pronumerals in the denominator or binomial numerators
- expand and factorise algebraic expressions

- (PATH) factorise algebraic expressions involving terms with indices
- expand binomial products and (PATH) special binomial products
- (PATH) factorise (monic) quadratic expressions of the form $x^2 + bx + c$
- (PATH) factorise an expression with 4 terms by grouping in pairs
- (PATH) factorise (non-monic) quadratic expressions of the form $ax^2 + bx + c$, including those that are perfect squares or the difference of 2 squares
- (PATH) factorise and simplify algebraic fractions
- (PATH) simplify, add, subtract, multiply and divide surds
- (PATH) expand and simplify binomial products involving surds
- (PATH) rationalise the denominator of expressions of the form $\frac{a\sqrt{b}}{c\sqrt{d}}$

SkillCheck ANSWERS ON P. 617

Quiz
SkillCheck 4

1 Simplify each expression.

a	$g^4 \times g^5$	**b**	$r^8 \div r^2$	**c**	$(d^5)^3$	**d**	$(-k)^2$
e	$h \times h^9$	**f**	$m^5 \div m$	**g**	a^1	**h**	c^0
i	$3e^2 \times 2e^5$	**j**	$18n^6 \div 6n^2$	**k**	$(10w^3)^3$	**l**	$25q^0$
m	$(vw)^5$	**n**	$\left(\frac{m}{n}\right)^3$	**o**	y^{-1}	**p**	p^{-2}

2 Simplify each expression.

a $\frac{a}{5} + \frac{3a}{4}$ **b** $\frac{7p}{2} - \frac{10p}{3}$ **c** $\frac{8}{t} \times \frac{5t}{24}$ **d** $\frac{x}{14} \div \frac{y}{2}$

3 Find 2 numbers whose:

a product is 18 and sum is 9 **b** product is 8 and sum is −6

c product is −20 and sum is −1 **d** product is −16 and sum is 6

4 Select the square numbers from this list:

44 81 25 100 75 72 16 50 64 32

The index laws 4.01

The index laws

$a^m \times a^n = a^{m+n}$ $a^m \div a^n = \frac{a^m}{a^n} = a^{m-n}$ $(a^m)^n = a^{m \times n}$

$(ab)^n = a^n b^n$ $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$ $a^0 = 1$

$a^{-n} = \frac{1}{a^n}$ $a^{-1} = \frac{1}{a}$ $\left(\frac{a}{b}\right)^{-1} = \frac{b}{a}$ $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n = \frac{b^n}{a^n}$

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Videos
Simplifying with the index laws
Index laws
Numbers and powers
Negative indices

Worksheet
Index laws review

Puzzles
Indices puzzle
Indices squaresaw

Presentation
Index laws

Example 1

Simplify each expression.

a $\dfrac{24e^6n^{12}}{8en^4}$ **b** $(4d^4q^2r)^3$ **c** $\left(\dfrac{2c^2}{d}\right)^4$

d $31x^0 - (31x)^0$ **e** $9x^{-1}$ **f** $(4q)^{-3}$

Solution

a $\dfrac{24e^6n^{12}}{8en^4} = \dfrac{{}^{3}\cancel{24}e^6n^{12}}{{}_{1}\cancel{8}en^4}$
$= 3e^5n^8$

b $(4d^4q^2r)^3 = 4^3d^{4\times3}q^{2\times3}r^3$
$= 64d^{12}q^6r^3$

c $\left(\dfrac{2c^2}{d}\right)^4 = \dfrac{(2c^2)^4}{d^4}$
$= \dfrac{2^4c^{2\times4}}{d^4}$
$= \dfrac{16c^8}{d^4}$

d $31x^0 - (31x)^0 = 31 \times 1 - 1$
$= 30$

e $9x^{-1} = 9 \times \dfrac{1}{x}$
$= \dfrac{9}{x}$

f $(4q)^{-3} = \dfrac{1}{(4q)^3}$
$= \dfrac{1}{64q^3}$

Example 2

Simplify:

a $\left(-1\frac{2}{3}\right)^{-5}$ **b** $\left(\dfrac{2}{3r}\right)^{-3}$

Solution

a $\left(-1\frac{2}{3}\right)^{-5} = \left(-\frac{5}{3}\right)^{-5}$
$= \left(-\frac{3}{5}\right)^{5}$
$= -\frac{243}{3125}$

b $\left(\dfrac{2}{3r}\right)^{-3} = \left(\dfrac{3r}{2}\right)^{3}$
$= \dfrac{27r^3}{8}$

EXERCISE 4.01 ANSWERS ON P. 618

The index laws U F R C

EXAMPLE 1

1 Simplify each expression. R C

a $8m^5 \times 3m^9$ **b** $30k^{10} \div 25k^2$ **c** $(-y^2)^5$

d $(3q^3)^2$ **e** $32a^7d^4 \div 8ad^3$ **f** $3x^6y^8 \times 7xy^4$

g $5e^4g^3 \times (-e^6g)$ **h** $\dfrac{27w^3y^8}{-9w^2y}$ **i** $(4c^7)^3$

j $(-2p)^5$ **k** $\dfrac{45hk^5m^3}{30hk^3m^2}$ **l** $9u^3vw^2 \times 6uv^2w^8$

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2 Simplify each expression. R C

a $(l^3m^5)^6$ **b** $\left(\frac{n}{2}\right)^3$ **c** $\left(\frac{w^4}{m^3}\right)^5$ **d** $\left(\frac{5}{2}\right)^0$

e $(-8ky^5)^2$ **f** $(3p^2q^3r^4)^4$ **g** $5x^0 - (7x)^0$ **h** $(-5d^3y^5)^3$

i $\left(\frac{2b}{3d}\right)^4$ **j** $-9(a^2b^3)^0$ **k** $12h^0$ **l** $\left(-\frac{3k^4}{10}\right)^3$

3 Evaluate each expression. R C

a 7^0 **b** 5×4^0 **c** $(-4)^3$ **d** $(2^3)^2$

e $6^5 \div 6^3$ **f** $(-3^2)^0$ **g** $8^2 \div 8^3$ **h** $5^4 \times 5^{-3}$

i $\left(\frac{3}{5}\right)^0$ **j** $7^5 \div 7^3$ **k** $12^2 \div 12^2$ **l** $3^5 \times 3^{-5}$

m $15^0 \div 15^1$ **n** $(4^{-1})^2$ **o** $(2^3)^0 + (2^0)^3$ **p** $(5^0)^3 \times 5^2$

4 Find the value of $(3x)^0 + 3 \times 2^0$. Select the correct answer **A**, **B**, **C** or **D**.

A 6 **B** 1 **C** 7 **D** 4

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5 Express each term as a fraction. R C

a 2^{-3} **b** 10^{-6} **c** 8^{-1} **d** 6^{-2} **e** 3^{-4}

6 Simplify each expression using a positive index. R C

a 5^{-4} **b** 8^{-2} **c** w^{-6} **d** m^{-5}

e $4y^{-3}$ **f** $(ad)^{-2}$ **g** $(2x)^{-5}$ **h** $3mp^{-2}$

i $10g^{-2}h^{-3}$ **j** $(4u)^{-4}$ **k** $2a^5q^{-2}$ **l** $4p^2x^{-3}$

m $10a^{-5}c^3$ **n** $(-3r^6)^{-3}$ **o** $\frac{3}{4}q^{-2}p^{-1}$ **p** $4(5k^3)^{-3}$

7 Simplify $2c^{-3}$. Select **A**, **B**, **C** or **D**.

A $-6c$ **B** $\frac{1}{2c^3}$ **C** $\frac{2}{c^3}$ **D** $\frac{1}{8c^3}$

8 Simplify each expression using positive indices. R C

a $\left(\frac{5}{2d}\right)^{-2}$ **b** $5y^{-2}$ **c** $(5m)^{-3}$ **d** $x^2y^3w^{-2}$

e $\left(\frac{4y^3}{3}\right)^{-2}$ **f** $\left(\frac{a}{7k}\right)^{-1}$ **g** $8(3ab^2)^{-2}$ **h** $\left(\frac{6c}{5k^2}\right)^{-1}$

9 Simplify each expression. R C

EXAMPLE 2

a $\left(\frac{1}{4}\right)^{-2}$ **b** $\left(\frac{5}{3}\right)^{-3}$ **c** $\left(-1\frac{2}{3}\right)^{-4}$ **d** $\left(2\frac{3}{5}\right)^{-2}$

e $\left(\frac{4}{m}\right)^{-3}$ **f** $\left(\frac{3a}{7}\right)^{-2}$ **g** $\left(-\frac{2}{y}\right)^{-5}$ **h** $\left(\frac{8q^2}{5}\right)^{-2}$

i $\left(-\frac{4}{w^3}\right)^{-3}$ **j** $\left(\frac{k^2}{d^5}\right)^{-2}$ **k** $\left(-\frac{10g^4}{3d^3}\right)^{-5}$ **l** $\left(\frac{7h^6}{5}\right)^{-4}$

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10 Simplify each expression. R C

a $(10x^{10}y)^3 \times 5x^{-2}y^3$

b $(20k^7m^2 \div 5k^{-4}m)^4$

c $(2q^{-3}r^3)^4 \div 8(qr^2)^2$

d $8w^{-5}m^3 \div (6wm^2)^2$

e $\left(\frac{4a^5x^6}{a^6x^2}\right)^4$

f $\left(\frac{3g^7n^6}{2g^5n^2}\right)^{-2}$

g $(4p^{-3}h^4)^3 \times (-2p^6h^9)$

h $(8d^7n^5)^2 \div (-2d^3n^2)^3$

i $(-2c^3e \times 5c^4e^5)^3$

4.02 Fractional indices

PATH

Puzzles
Indices squaresaw (Advanced)

Exponential equations

Fractional indices

$a^{\frac{1}{2}} = \sqrt{a}$ Any number raised to the power of $\frac{1}{2}$ is the square root of that number.

$a^{\frac{1}{3}} = \sqrt[3]{a}$ Any number raised to the power of $\frac{1}{3}$ is the cube root of that number.

$a^{\frac{1}{n}} = \sqrt[n]{a}$ Any number raised to the power of $\frac{1}{n}$ is the nth root of that number.

$a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m$ or $\sqrt[n]{a^m}$ Any number raised to the power of $\frac{m}{n}$ is the nth root of that number raised to the power of m.

Note: Taking the root first often makes the calculation simpler.

On a calculator, the nth root key is $\sqrt{\square}$ or $\sqrt[x]{\square}$ found by pressing the SHIFT or 2nd F key before pressing $x^{\square}$ or y^x respectively.

Example 3

Evaluate each expression.

a $961^{\frac{1}{2}}$ **b** $1331^{\frac{1}{3}}$ **c** $6561^{\frac{1}{8}}$

Solution

a $961^{\frac{1}{2}} = \sqrt{961}$
$= 31$

b $1331^{\frac{1}{3}} = \sqrt[3]{1331}$
$= 11$

c $6561^{\frac{1}{8}} = \sqrt[8]{6561}$
$= 3$

On a calculator: 8 $\sqrt{\square}$ 6561 =
because $3^8 = 6561$

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Example 4

Evaluate each expression.

a $32^{\frac{3}{5}}$ **b** $27^{-\frac{1}{3}}$ **c** $25^{-\frac{5}{2}}$

Solution

a $32^{\frac{3}{5}} = \left(\sqrt[5]{32}\right)^3$

$= 2^3$

$= 8$

b $27^{-\frac{1}{3}} = \frac{1}{27^{\frac{1}{3}}}$

$= \frac{1}{\sqrt[3]{27}}$

$= \frac{1}{3}$

c $25^{-\frac{5}{2}} = \frac{1}{25^{\frac{5}{2}}}$

$= \frac{1}{\left(\sqrt{25}\right)^5}$

$= \frac{1}{5^5}$

$= \frac{1}{3125}$

Example 5

Write each expression using a fractional index.

a $\sqrt{w^9}$ **b** $\sqrt[4]{m^5}$ **c** $\frac{1}{\sqrt[3]{y^8}}$

Solution

a $\sqrt{w^9} = \left(w^9\right)^{\frac{1}{2}}$

$= w^{\frac{9}{2}}$

b $\sqrt[4]{m^5} = \left(m^5\right)^{\frac{1}{4}}$

$= m^{\frac{5}{4}}$

c $\frac{1}{\sqrt[3]{y^8}} = \frac{1}{\left(y^8\right)^{\frac{1}{3}}}$

$= \frac{1}{y^{\frac{8}{3}}}$ or $y^{-\frac{8}{3}}$

Example 6

Simplify each expression.

a $\sqrt[4]{\left(81a^2\right)^5}$ **b** $\left(64k^6\right)^{-\frac{5}{3}}$ **c** $4m^{\frac{3}{4}}n^2 \times \sqrt[3]{8m^2n}$

Solution

a $\sqrt[4]{\left(81a^2\right)^5} = \left(81a^2\right)^{\frac{5}{4}}$

$= 81^{\frac{5}{4}} a^{2\times\frac{5}{4}}$

$= 243a^{\frac{5}{2}}$

b $\left(64k^6\right)^{-\frac{5}{3}} = \frac{1}{\left(64k^6\right)^{\frac{5}{3}}}$

$= \frac{1}{64^{\frac{5}{3}} k^{6\times\frac{5}{3}}}$

$= \frac{1}{1024k^{10}}$

c $4m^{\frac{3}{4}}n^2 \times \sqrt[3]{8m^2n} = 4m^{\frac{3}{4}}n^2 \times \left(8m^2n\right)^{\frac{1}{3}}$

$= 4m^{\frac{3}{4}}n^2 \times 8^{\frac{1}{3}} m^{2\times\frac{1}{3}} n^{\frac{1}{3}}$

$= 4m^{\frac{3}{4}}n^2 \times 2m^{\frac{2}{3}} n^{\frac{1}{3}}$

$= 8m^{\frac{3}{4}+\frac{2}{3}} n^{2+\frac{1}{3}}$

$= 8m^{\frac{17}{12}} n^{\frac{7}{3}}$

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Video
Fractional indices

4.02

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EXERCISE 4.02 ANSWERS ON P. 618

Fractional indices U F R C

EXAMPLE 3

1 Evaluate each expression. R

a $64^{\frac{1}{2}}$ **b** $-27^{\frac{1}{3}}$ **c** $625^{\frac{1}{2}}$ **d** $1000^{\frac{1}{3}}$

e $16^{\frac{1}{4}}$ **f** $(-0.00032)^{\frac{1}{5}}$ **g** $(0.01)^{\frac{1}{2}}$ **h** $(-512)^{\frac{1}{9}}$

i $(-8)^{\frac{1}{3}}$ **j** $(-729)^{\frac{1}{3}}$ **k** $256^{\frac{1}{8}}$ **l** $3125^{\frac{1}{5}}$

2 Write each expression using a radical (root) sign. R C

a $7^{\frac{1}{3}}$ **b** $15^{\frac{1}{2}}$ **c** $d^{\frac{1}{4}}$ **d** $y^{\frac{1}{2}}$

e $(4x)^{\frac{1}{6}}$ **f** $(10m^3)^{\frac{1}{2}}$ **g** $(2a^7)^{\frac{1}{10}}$ **h** $(12m)^{\frac{1}{3}}$

3 Write $\sqrt[4]{100k}$ using a fractional index. Select the correct answer **A**, **B**, **C** or **D**. R C

A $25k^{\frac{1}{4}}$ **B** $100k^{\frac{1}{4}}$ **C** $(100k)^{\frac{1}{4}}$ **D** $5k^{\frac{1}{4}}$

4 Write each expression using a fractional index. R C

a $\sqrt[3]{80}$ **b** $\sqrt[6]{40}$ **c** $\sqrt{20}$ **d** $\sqrt[4]{c}$

e $\sqrt{b}$ **f** $\sqrt[3]{xy}$ **g** $\sqrt[5]{25w}$ **h** $\sqrt[8]{4y}$

EXAMPLE 4

5 Evaluate each expression. R C

a $16^{\frac{3}{4}}$ **b** $25^{\frac{5}{2}}$ **c** $(-27)^{-\frac{5}{3}}$ **d** $8^{\frac{2}{3}}$

e $(-128)^{-\frac{3}{7}}$ **f** $32^{-\frac{1}{5}}$ **g** $1000^{\frac{5}{3}}$ **h** $900^{-\frac{1}{2}}$

i $64^{\frac{5}{6}}$ **j** $9^{-\frac{7}{2}}$ **k** $(-216)^{-\frac{4}{3}}$ **l** $(-243)^{-\frac{2}{5}}$

6 Evaluate the expression $100\,000^{-\frac{7}{5}}$. Select **A**, **B**, **C** or **D**. R C

A $\frac{1}{1\,000\,000}$ **B** $\frac{1}{10\,000\,000}$ **C** $\frac{1}{1000}$ **D** $\frac{1}{100\,000}$

7 Evaluate each expression correct to 2 decimal places.

a $\sqrt[3]{-250}$ **b** $37^{\frac{1}{2}}$ **c** $14^{-\frac{3}{4}}$ **d** $\sqrt[7]{-138}$

e $4^{-\frac{5}{3}}$ **f** $200^{\frac{3}{10}}$ **g** $(-50)^{-\frac{6}{5}}$ **h** $\sqrt[4]{82}$

EXAMPLE 5

8 Write each expression using a fractional index. R C

a $\sqrt[4]{k^3}$ **b** $\frac{1}{\sqrt{m}}$ **c** $\sqrt[3]{x^7}$ **d** $\sqrt{y^3}$

e $\frac{1}{\sqrt[5]{d^2}}$ **f** $\sqrt{2a^3}$ **g** $\frac{1}{\sqrt[5]{4p}}$ **h** $\frac{1}{\sqrt[4]{vw}}$

EXAMPLE 6

9 Simplify $\sqrt[3]{(8d^2)^5}$. Select **A**, **B**, **C** or **D**. R C

A $32d^{30}$ **B** $4d^{\frac{5}{3}}$ **C** $16d^{\frac{10}{3}}$ **D** $32d^{\frac{10}{3}}$



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10 Simplify each expression. R C

a $\left(25m^8\right)^{\frac{5}{2}}$ **b** $\sqrt[3]{\left(8y^6\right)^2}$ **c** $\sqrt[4]{\left(16x^2w^6\right)^6}$

d $\left(27c^6\right)^{-\frac{5}{3}}$ **e** $\left(8k^9\right)^{-\frac{4}{3}}$ **f** $\left(\frac{1}{\sqrt[5]{32p^{15}}}\right)^2$

g $\left(10\,000n^4w^6\right)^{-\frac{1}{2}}$ **h** $\sqrt[5]{\left(243x^{10}y^5\right)^4}$ **i** $6p^{\frac{2}{3}}n^{\frac{1}{2}} \times \sqrt{36p^4n^5}$

j $\frac{1}{\left(\sqrt[4]{625y^8d^4}\right)^3}$ **k** $\left(64m^6\right)^{-\frac{1}{3}} \times (4m)^2$ **l** $\sqrt[3]{\left(216a^6b^9\right)^2}$

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4.03

Adding and subtracting algebraic fractions

4.03

Example 7

Simplify each expression.

a $\frac{a}{2}+\frac{a}{3}$ **b** $\frac{2x}{5}-\frac{x}{3}$ **c** $\frac{5}{y}+\frac{7}{2y}$

Solution

a
$$\frac{a}{2}+\frac{a}{3}=\frac{3\times a}{3\times 2}+\frac{2\times a}{2\times 3}$$
$$=\frac{3a}{6}+\frac{2a}{6}$$
$$=\frac{5a}{6}$$

b
$$\frac{2x}{5}-\frac{x}{3}=\frac{3\times 2x}{3\times 5}-\frac{5\times x}{5\times 3}$$
$$=\frac{6x}{15}-\frac{5x}{15}$$
$$=\frac{x}{15}$$

c
$$\frac{5}{y}+\frac{7}{2y}=\frac{2\times 5}{2y}+\frac{7}{2y}$$
$$=\frac{10}{2y}+\frac{7}{2y}$$
$$=\frac{17}{2y}$$

Videos
Adding and subtracting algebraic fractions
Algebraic fractions

Example 8

Simplify each expression.

a $\frac{6h+1}{8}+\frac{4h-7}{12}$ **b** $\frac{x}{3}-\frac{x+1}{5}$

Solution

a
$$\frac{6h+1}{8}+\frac{4h-7}{12}=\frac{3(6h+1)}{24}+\frac{2(4h-7)}{24}$$
$$=\frac{18h+3}{24}+\frac{8h-14}{24}$$
$$=\frac{18h+3+8h-14}{24}$$
$$=\frac{26h-11}{24}$$

b
$$\frac{x}{3}-\frac{x+1}{5}=\frac{5x}{15}-\frac{3(x+1)}{15}$$
$$=\frac{5x}{15}-\frac{3x+3}{15}$$
$$=\frac{5x-(3x+3)}{15}$$
$$=\frac{5x-3x-3}{15}$$
$$=\frac{2x-3}{15}$$

PATH

Video
Algebraic fractions

Puzzle
Algebraic fractions

Foundation Standard Complex

EXERCISE 4.03 ANSWERS ON P. 618

Adding and subtracting algebraic fractions U F

EXAMPLE 7

1 Simplify each expression.

a $\frac{n}{2}-\frac{n}{7}$ **b** $\frac{3c}{2}+\frac{c}{5}$ **c** $\frac{4r}{7}+\frac{5r}{2}$ **d** $\frac{7y}{8}-\frac{5y}{3}$

e $\frac{4t}{3}+\frac{t}{9}$ **f** $\frac{5y}{16}-\frac{3y}{8}$ **g** $\frac{11t}{12}-\frac{5t}{9}$ **h** $\frac{3a}{10}+\frac{4a}{15}$

PATH

i $\frac{4}{k}+\frac{3a}{k}$ **j** $\frac{7d}{3y}-\frac{2d}{3y}$ **k** $\frac{12}{5x}-\frac{7}{5x}$ **l** $\frac{11p}{12}+\frac{3p}{12}$

m $\frac{5}{q}+\frac{3}{r}$ **n** $\frac{15}{4h}-\frac{2}{h}$ **o** $\frac{5}{3b}+\frac{3}{4b}$ **p** $\frac{9}{2n}-\frac{4}{3p}$

2 Simplify each expression.

a $\frac{w}{7}+\frac{n}{4}$ **b** $\frac{3x}{8}+\frac{5u}{7}$ **c** $\frac{2a}{15}-\frac{3h}{10}$ **d** $\frac{d}{16}-\frac{r}{24}$

e $\frac{4k}{5}-\frac{g}{9}$ **f** $\frac{5m}{12}+\frac{2r}{5}$ **g** $\frac{3h}{2}+\frac{4}{5}$ **h** $\frac{5u}{4}-\frac{3a}{7}$

i $\frac{a}{5}+\frac{c}{9}$ **j** $\frac{5q}{12}-\frac{11w}{18}$ **k** $\frac{5k}{8}+\frac{3}{11}$ **l** $\frac{4n}{9}+\frac{5n}{6}$

3 Simplify $\frac{5p}{6}-\frac{3p}{8}$. Select the correct answer **A**, **B**, **C** or **D**.

A $\frac{29p}{24}$ **B** $\frac{22p}{48}$ **C** $\frac{11p}{24}$ **D** $\frac{2p}{24}$

PATH

4 Simplify $\frac{1}{xw}+\frac{w}{xy}$. Select **A**, **B**, **C** or **D**.

A $\frac{y+w}{xyw}$ **B** $\frac{w^2+y}{xyw}$ **C** $\frac{y+w^2}{xy}$ **D** $\frac{y^2+w}{wx}$

5 Simplify $\frac{4w-3}{5}-\frac{1+2w}{4}$. Select **A**, **B**, **C** or **D**.

EXAMPLE 8

A $\frac{6w-17}{20}$ **B** $\frac{6w-7}{20}$ **C** $\frac{26w-17}{20}$ **D** $\frac{26w-7}{10}$

6 Simplify each expression.

a $\frac{m+3}{5}+\frac{m-2}{4}$ **b** $\frac{3p}{5}+\frac{p+2}{3}$ **c** $\frac{y+3}{4}+\frac{2y-1}{3}$

d $\frac{x-1}{7}+\frac{x-2}{3}$ **e** $\frac{5h+6}{2}-\frac{2h+2}{9}$ **f** $\frac{k+3}{10}-\frac{2k-3}{7}$

g $\frac{a-1}{2}-\frac{a}{3}$ **h** $\frac{3d-1}{7}+\frac{2d-9}{6}$ **i** $\frac{2y+5}{4}-\frac{y-1}{9}$

j $\frac{1-w}{10}-\frac{3w-2}{7}$ **k** $\frac{9-2e}{3}-\frac{e-5}{2}$ **l** $\frac{8-3x}{11}-\frac{2x+5}{4}$

Foundation Standard Complex

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Multiplying and dividing algebraic fractions

4.04

Example 9

Simplify each expression.

a $\frac{3x}{4} \times \frac{2x}{9}$ **b** $\frac{4}{k} \times \frac{3k}{16}$ **c** $\frac{2}{v} \div \frac{3}{w}$ **d** $\frac{xy}{5} \div \frac{3x}{25}$

Solution

a $\frac{3x}{4} \times \frac{2x}{9} = \frac{3x \times 2x}{4 \times 9}$

$= \frac{6x^2}{36}$

$= \frac{x^2}{6}$

or $\frac{3x}{4} \times \frac{2x}{9} = \frac{{}^{1}\cancel{3}x}{{}_{2}\cancel{4}} \times \frac{{}^{1}\cancel{2}x}{{}_{3}\cancel{9}}$

$= \frac{x^2}{6}$

b $\frac{4}{k} \times \frac{3k}{16} = \frac{{}^{1}\cancel{4}}{\cancel{k}} \times \frac{3\cancel{k}}{{}_{4}\cancel{16}}$

$= \frac{3}{4}$

c $\frac{2}{v} \div \frac{3}{w} = \frac{2}{v} \times \frac{w}{3}$

$= \frac{2w}{3v}$

d $\frac{xy}{5} \div \frac{3x}{25} = \frac{\cancel{x}y}{{}_{1}\cancel{5}} \times \frac{{}^{5}\cancel{25}}{3\cancel{x}}$

$= \frac{5y}{3}$

Worksheet
Algebraic fractions

Puzzles
Algebraic fractions
Upside-down fractions

Video
Algebraic fractions

4.04

EXERCISE 4.04 ANSWERS ON P. 618

Multiplying and dividing algebraic fractions U F

EXAMPLE 9

PATH

1 Simplify each product.

a $\frac{5}{6} \times \frac{d}{7}$ **b** $\frac{x}{3} \times \frac{w}{8}$ **c** $\frac{k}{5} \times \frac{b}{3}$

d $\frac{3h}{5} \times \frac{m}{4}$ **e** $\frac{w}{15} \times \frac{10y}{2}$ **f** $\frac{4a}{3} \times \frac{5c}{12}$

g $\frac{2x}{3} \times \frac{9}{8y}$ **h** $\frac{6}{5b} \times \frac{10b}{18}$ **i** $\frac{8h}{p} \times \frac{2h}{3g}$

j $\frac{5u}{12c} \times \frac{8c}{15u}$ **k** $\frac{9w}{45pq} \times \frac{5p}{6wh}$ **l** $\frac{20a}{3k} \times \frac{9a}{5k}$

2 Simplify each quotient.

a $\frac{m}{3} \div \frac{g}{7}$ **b** $\frac{xy}{24} \div \frac{5x}{6}$ **c** $\frac{4}{5} \div \frac{5d}{2}$

d $\frac{8a}{3} \div \frac{5a}{9}$ **e** $\frac{w}{15} \div \frac{w}{5}$ **f** $\frac{9y}{4} \div \frac{5y}{12}$

g $\frac{2c^2}{5e} \div \frac{8c}{3}$ **h** $\frac{10g}{7kc} \div \frac{2g}{21k}$ **i** $\frac{4}{3xy} \div \frac{7k}{9y}$

j $\frac{5d^2}{6} \div \frac{d}{10}$ **k** $\frac{4e}{15p^2} \div \frac{6e^2}{5p}$ **l** $\frac{9r}{5} \div \frac{12r^2}{35g}$

PATH

3 Simplify the quotient $\frac{5xy}{3k} \div \frac{20}{9ky}$. Select the correct answer **A**, **B**, **C** or **D**.

A $\frac{3y}{4x}$ **B** $\frac{4x}{15y^2}$ **C** $\frac{45k}{60y}$ **D** $\frac{3xy^2}{4}$

Foundation Standard Complex

PATH

4 Simplify the product $\frac{6ad}{7m} \times \frac{3y}{4d} \times \frac{14gm}{27ay}$. Select the answer **A**, **B**, **C** or **D**.

A $\frac{2ak}{7y}$ **B** $\frac{g}{3}$ **C** $\frac{14gm}{9}$ **D** $\frac{7m}{4}$

5 Simplify each expression.

a $\frac{8}{xy} \div \frac{6}{xw}$ **b** $\frac{3dg}{5k} \div \frac{9kg}{4h} \times \frac{6k}{5h}$ **c** $\frac{12mn}{5} \div \frac{8m}{15}$

d $\frac{10a}{3pf} \div \frac{5}{21p}$ **e** $\frac{8c}{5} \div 4$ **f** $\frac{3n}{v} \div \frac{5}{4v} \div \frac{9nm}{10}$

g $\frac{4ab}{15d} \div \frac{3d}{8b} \times \frac{5bd}{16}$ **h** $24hw \div \frac{4kw}{9y}$ **i** $\frac{8dm}{3} \div \frac{m}{6} \times \frac{1}{12d^2}$

j $\frac{10}{3p} \times \frac{pq}{5h} \div \frac{6q}{5p}$ **k** $\frac{8ac}{w} \div \frac{4w}{9c} \div \frac{2c}{yh}$ **l** $\frac{2u}{7y} \div \frac{3}{4xy} \times \frac{21y}{8x}$

4.05 Expanding and factorising expressions

Skillsheets
Algebra using diagrams
HCF by factor trees
Factorising using diagrams

Worksheets
Expanding and factorising
Algebra 4

Puzzles
The distributive law
Factorising puzzle

Video
Factorising expressions 2

Example 10

Expand $-3r^2(4r + 2) - 5r^3$.

Solution

$-3r^2(4r + 2) - 5r^3 = -3r^2 \times 4r + (-3r^2) \times 2 - 5r^3$ Expanding

$= -12r^3 + (-6r^2) - 5r^3$

$= -17r^3 - 6r^2$ Collecting like terms to simplify.

Example 11

Factorise each expression.

a $3m(5 + 2d) + 7(5 + 2d)$ **b** $-5k^2 + 15k$

Solution

a The **HCF** of $3m(5 + 2d) + 7(5 + 2d)$ is $(5 + 2d)$.

$\therefore 3m(5 + 2d) + 7(5 + 2d) = (5 + 2d) \times 3m + (5 + 2d) \times 7$

$= (5 + 2d)(3m + 7)$

b When factorising expressions that begin with a negative term, we use the 'negative' HCF.

The highest 'negative' common factor of $-5k^2$ and $15k$ is $-5k$.

$\therefore -5k^2 + 15k = (-5k) \times k + (-5k) \times (-3)$ $(-5k) \times (-3) = +15k$

$= (-5k)[k + (-3)]$

$= -5k(k - 3)$

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Example 12

Factorise each expression.

a $8a^3 + 4a^2$ **b** $20h^3k + 25h^4k - 10h^2k$

Solution

a The HCF of $8a^3$ and $4a^2$ is $4a^2$.

$\therefore 8a^3 + 4a^2 = 4a^2 \times 2a + 4a^2 \times 1$ Rewrite the expression using the HCF $4a^2$.

$= 4a^2(2a + 1)$

b The HCF is $5h^2k$.

$\therefore 20h^3k + 25h^4k - 10h^2k = 5h^2k \times 4h + 5h^2k \times 5h^2 - 5h^2k \times 2.$

$= 5h^2k(4h + 5h^2 - 2)$

EXERCISE 4.05 ANSWERS ON P. 619

Expanding and factorising expressions U F

1 Expand each expression.

a	$5(d + 11)$	**b**	$-3(r + 10)$	**c**	$7(x - 9y)$	**d**	$-4(a - 5w)$
e	$-(2 - p^2)$	**f**	$-10e(2e^2 + 3)$	**g**	$6y(1 + 7y)$	**h**	$4xy(3xy - 1)$
i	$8rq(2q - r)$	**j**	$3ab(4b - 7a)$	**k**	$-6h^2(1 - 3h)$	**l**	$-5x(5x^2 + 4y)$
m	$-(3 + 8a)$	**n**	$-2m^2(3m - 4n)$	**o**	$5g(3 + 7g^2)$	**p**	$-1(5e - 12)$

EXAMPLE 10

2 Expand and simplify by collecting like terms.

a	$4k(3k - 5) - 9k^2$	**b**	$5h - 7h(4 - h)$
c	$9w^3 - 3w(5 + 2w^2)$	**d**	$24x^3 - 5x^2(2x^2 - 5x)$
e	$8 - 3(2 - 7d)$	**f**	$4n(3 - 5n) - 6n^2$
g	$4y(y - 4) + 5(2y + 1)$	**h**	$5(1 - 2a) - a(3 + 4a)$
i	$7(3 + 6w) - (5 - 8w)$	**j**	$2c(5c - 1) - 4(7c - 5)$
k	$3m(m + 5m^2) - m^2(1 - 3m)$	**l**	$4x(2y + 5) - 6y(10 - 3x)$

EXAMPLE 11

3 Factorise each expression.

a	$28k^2 - 21k$	**b**	$8(c - 5) - c(c - 5)$	**c**	$m(3 + 2m) + 7(3 + 2m)$
d	$-q^2 - 36q$	**e**	$-8x + 12x^2$	**f**	$b(3b + 5) - 2(3b + 5)$
g	$-12cd^2 + 8cd$	**h**	$-hn^2 + h^2n$	**i**	$-15g^2 - 18g$

4 Factorise $-16pw + 10xw$. Select the correct answer **A**, **B**, **C** or **D**.

A $-2w(8p - 10x)$ **B** $-2w(8p - 5x)$ **C** $-8w(2p - x)$ **D** $-4w(5p + 2x)$

PATH

EXAMPLE 12

5 Factorise each expression.

a $8m^2y^2 - 12my$

b $36ab^2c + 27bc$

c $24m^2n - 108mn^2$

d $20dg^2 - 35ag$

e $40wy^3 + 24w^2y^2$

f $75g^3h^2 - 125gh$

g $-4p^3 - 8p^2 + p$

h $6mn^2 + 3mn + 48m^2n$

i $32p^3g + 8pg^2 - 8pg$

j $18a^5 - 12a^2 + 15a^4$

k $28m^3h^2 - 21mh^2$

l $15kwp - 24wp^2 - 9kw$

Did you know?

CAPTCHA

For security purposes on a website, have you ever been asked to enter letters, words or numbers that have been displayed in a wavy, difficult-to-read format like this?



This process is called **CAPTCHA**, which stands for **Completely Automated Public Turing Test to Tell Computers and Humans Apart**, invented in 1999. It is designed to ensure that the person accessing the website is a human and not another computer that could be hacking into the site, sending spam or viruses. CAPTCHA uses optical character recognition (OCR), which is a technology that can convert images of text into editable text.

How does CAPTCHA prove that you are a human and not a computer?

What is a Turing test, named after English computer scientist and mathematician Alan Turing?

Expanding binomial products

4.06

Binomial means '2 terms'. $(m + 8)(m - 3)$ is called a **binomial product** because it is a product of 2 **binomial expressions**.

Example 13

Expand each binomial product.

a $(m + 8)(m - 3)$ **b** $(4y - 5)(3y - 2)$

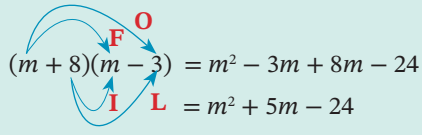
Solution

a $(m + 8)(m - 3) = m(m - 3) + 8(m - 3)$ Each term in $(m + 8)$ is multiplied by $(m - 3)$

$= m^2 - 3m + 8m - 24$ Expanding

$= m^2 + 5m - 24$ Simplifying

One way of remembering which pairs of terms to multiply together in a binomial product is called the **FOIL method** (First-Outer-Inner-Last), as shown.



b $(4y - 5)(3y - 2) = 12y^2 - 8y - 15y + 10$ Using FOIL

$= 12y^2 - 23y + 10$ Simplifying

Perfect squares

$$(a + b)^2 = a^2 + 2ab + b^2$$
$$(a - b)^2 = a^2 - 2ab + b^2$$

Example 14

Expand each perfect square.

a $(n - 5)^2$ **b** $(k + 4)^2$ **c** $(3y - 8)^2$

Solution

a $(n - 5)^2 = n^2 - 2 \times n \times 5 + 5^2$ 1st term squared − double product + 2nd term squared

$= n^2 - 10n + 25$

b $(k + 4)^2 = k^2 + 2 \times k \times 4 + 4^2$

$= k^2 + 8k + 16$

c $(3y - 8)^2 = (3y)^2 - 2 \times 3y \times 8 + 8^2$

$= 9y^2 - 48y + 64$

Worksheets
Binomial products
Area diagrams
Mixed expansions
Algebra 2
Algebra 6

Puzzles
Expanding brackets
Trinominoes

Technology
Expanding binomials

Spreadsheet
Expanding binomials

PATH

Videos
Expanding binomial products 1
Expanding binomial products 2
Special binomial products

PATH

Videos
Difference of 2 squares

Special binomial products

Puzzles
Products and factors squaresaw

Factorominoes

Enough time

Worksheet
Special products

Difference of 2 squares

$$(a + b)(a - b) = a^2 - b^2$$

When the sum of 2 terms is multiplied by their difference, the answer is the square of the first term minus the square of the second term (the **difference of 2 squares**).

Example 15

Expand each expression.

a $(8g - 3)(8g + 3)$ **b** $(2x + 7w)(2x - 7w)$

Solution

a $(8g - 3)(8g + 3) = (8g)^2 - 3^2$
$= 64g^2 - 9$

b $(2x + 7w)(2x - 7w) = (2x)^2 - (7w)^2$
$= 4x^2 - 49w^2$

Example 16

Expand and simplify each expression.

a $x(5y - x)^2$ **b** $(2a - 3)(2a + 3) - (a + 3)^2$

Solution

a $x(5y - x)^2 = x(25y^2 - 10xy + x^2)$
$= 25xy^2 - 10x^2y + x^3$

b $(2a - 3)(2a + 3) - (a + 3)^2 = 4a^2 - 9 - (a^2 + 6a + 9)$
$= 4a^2 - 9 - a^2 - 6a - 9$
$= 3a^2 - 6a - 18$

EXERCISE 4.06 ANSWERS ON P. 619

Expanding binomial products U F R C

EXAMPLE 13

1 Expand $(5 + 3b)(2b - 7)$. Select the correct answer **A**, **B**, **C** or **D**.

A $6b^2 + 11b - 35$ **B** $6b^2 - 11b - 35$ **C** $6b^2 - 31b - 35$ **D** $11b - 6b^2 - 35$

2 Expand each binomial product.

a $(3m + 1)(m + 7)$ **b** $(4y + 3)(7y + 5)$ **c** $(1 + 4k)(k + 6)$

d $(5h - 2)(2h + 3)$ **e** $(v - 8)(2v - 9)$ **f** $(5a + 2)(4a - 3)$

g $(11 - 3e)(5 - 4e)$ **h** $(8 + g)(5 - 6g)$ **i** $(4y - 9)(7 - 2y)$

j $(2 - 3a)(4a + 3)$ **k** $(12 - 5k)(7 - 4k)$ **l** $(2w + 9)(7w - 10)$

Foundation Standard Complex

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PATH

4.06

3 Expand each perfect square.

EXAMPLE 14

a $(h+7)^2$ **b** $(p-5)^2$ **c** $(w-6)^2$ **d** $(4-c)^2$

e $(5u+1)^2$ **f** $(3b+7)^2$ **g** $(7e-11)^2$ **h** $(2-5h)^2$

i $(6m+5c)^2$ **j** $\left(2+\frac{2}{a}\right)^2$ **k** $\left(\frac{6}{m}-m\right)^2$ **l** $\left(\frac{3}{x}+3x\right)^2$

4 Expand $(4a-5w)^2$. Select **A**, **B**, **C** or **D**.

A $16a^2+25w^2$ **B** $16a^2-9aw-25w^2$

C $16a^2+18aw-25w^2$ **D** $16a^2-40aw+25w^2$

5 Expand each expression.

EXAMPLE 15

a $(k+7)(k-7)$ **b** $(12+w)(12-w)$ **c** $(4+d)(4-d)$

d $(7b-5)(7b+5)$ **e** $(3h+7)(3h-7)$ **f** $(9h-5m)(9h+5m)$

g $\left(\frac{4}{m}-m\right)\left(\frac{4}{m}+m\right)$ **h** $\left(3g+\frac{1}{g}\right)\left(3g-\frac{1}{g}\right)$ **i** $\left(\frac{4a}{3}+2\right)\left(\frac{4a}{3}-2\right)$

6 Expand $(8g-3y)(8g+3y)$. Select **A**, **B**, **C** or **D**.

A $16g^2-9y^2$ **B** $64g^2-9y^2$ **C** $16g^2-6w^2$ **D** $64g^2-48gy+25w^2$

7 Expand each expression.

a $(3t-d)^2$ **b** $(9-4g)(4g+9)$ **c** $(ab-8)(ab+8)$

d $(10x-7y)^2$ **e** $(2-11hm)(2+11hm)$ **f** $\left(xy-\frac{1}{y}\right)\left(xy+\frac{1}{y}\right)$

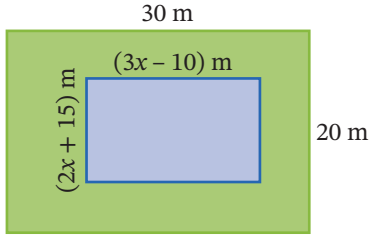
8 Expand and simplify each expression.

EXAMPLE 16

a $(3-y)^2-(3+y)^2$ **b** $8x^2-(2x-5)(2x+5)$

c $(m+2)^2-(m+2)(m-2)-(m-2)^2$ **d** $2(6-d)^2-(d+5)^2-(6-d)(d+5)$

9 This diagram shows a house $(3x-10)$ metres long and $(2x+15)$ metres wide on a block of land with dimensions 30 m × 20 m. R C



a Write down a binomial expression for the area of the house in square metres.

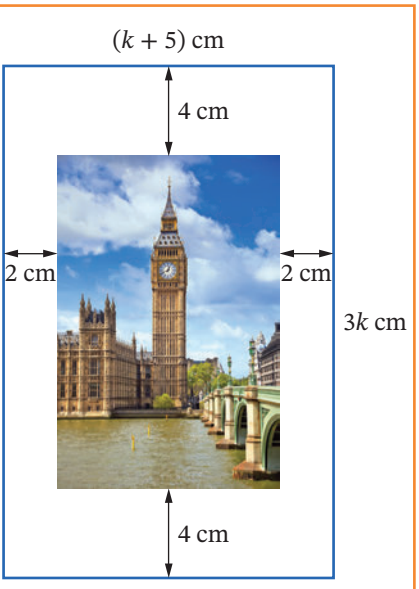
b Expand and simplify this expression.

c The green area of the block of land not covered by the house is to be turfed. Write a simplified expression for this area in square metres.

Foundation Standard Complex

PATH

10 A photograph frame is $(k + 5)$ cm wide and $3k$ cm long. The gap between the photograph and frame is 4 cm at the top and bottom and 2 cm on each side. R C



S.Borisov/Shutterstock.com

a What is the area of the photograph frame?

b Write down expressions for the length and width of the photograph.

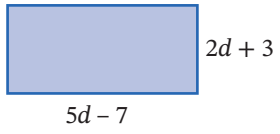
c Write down a binomial expression for the area of the photograph.

d Expand and simplify your expression for the area of the photograph.

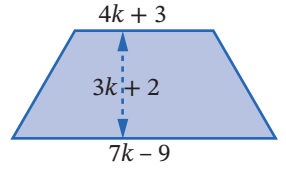
e Find an expression for the area of the frame not taken up by the photograph.

11 Find an expression for the area of each shape in simplest form. R C

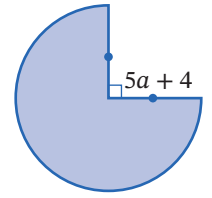
a



b



c



Investigation

Squaring a number ending in 5

Study this mental shortcut for squaring a number ending in 5:

- To evaluate 35^2, calculate $3 \times 4 = 12$, add '25' to the end: $35^2 = 1225$.
- To evaluate 75^2, calculate $7 \times 8 = 56$, add '25' to the end: $75^2 = 5625$.
- To evaluate 105^2, calculate $10 \times 11 = 110$, add '25' to the end: $105^2 = 11\,025$.

Let n stand for the tens digit of the number ending in 5 being squared.

Expand $(10n + 5)^2$ and investigate why the above method works.

Foundation Standard Complex

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Factorising special binomial products 4.07

An **algebraic expression with 4 terms** can often be factorised **by grouping in pairs**, that is, 2 terms at a time, to make a binomial product.

Example 17

Factorise each expression.

a $3xy - 4px - 6y^2 + 8py$ **b** $12aw + 20cx - 8cw - 30ax$

Solution

a
$$\begin{aligned} 3xy - 4px - 6y^2 + 8py &= x(3y - 4p) - 2y(3y - 4p) && \text{Factorising each pair} \\ &= (3y - 4p)(x - 2y) && \text{Factorising again} \end{aligned}$$

b
$$\begin{aligned} 12aw + 20cx - 8cw - 30ax &= 2(6aw + 10cx - 4cw - 15ax) && \text{Factorising all terms first} \\ &= 2(6aw - 15ax - 4cw + 10cx) && \text{Grouping into pairs} \\ &= 2[3a(2w - 5x) - 2c(2w - 5x)] && \text{Factorising each pair} \\ &= 2(2w - 5x)(3a - 2c) && \text{Factorising again} \end{aligned}$$

The difference of 2 squares

$$a^2 - b^2 = (a + b)(a - b)$$

Example 18

Factorise each expression.

a $m^2 - 25$ **b** $36 - 49y^2$ **c** $48k^2 - 3x^2$ **d** $w^4 - 4w^2$

Solution

a
$$\begin{aligned} m^2 - 25 &= m^2 - 5^2 \\ &= (m + 5)(m - 5) \end{aligned}$$

b
$$\begin{aligned} 36 - 49y^2 &= 6^2 - (7y)^2 \\ &= (6 + 7y)(6 - 7y) \end{aligned}$$

c
$$\begin{aligned} 48k^2 - 3x^2 &= 3(16k^2 - x^2) \\ &= 3[(4k)^2 - x^2] \\ &= 3(4k + x)(4k - x) \end{aligned}$$

d
$$\begin{aligned} w^4 - 4w^2 &= w^2(w^2 - 4) \\ &= w^2(w^2 - 2^2) \\ &= w^2(w + 2)(w - 2) \end{aligned}$$

PATH



Puzzle Grouping



Puzzle Difference of 2 perfect squares



Video Difference of 2 squares

PATH

EXERCISE 4.07 ANSWERS ON P. 619

Factorising special binomial products U F

EXAMPLE 17

1 Factorise each expression.

a $15mk + 20mg + 6nk + 8ng$

b $4yx - 8ay + 7ax - 14a^2$

c $14ak - 35af + 8k - 20f$

d $cd + cy - hd - hy$

e $4am + 4at + 4em + 4et$

f $3yk - 6by + 12k - 24b$

g $30y^2 - 6yq - 15y + 3q$

h $32gh + 24gk - 4mh - 3mk$

i $6ng - 10ch - 4cg + 15nh$

j $p^2w - p^3 - w + p$

k $m(d - 3) - mx + a(d - 3) - ax$

l $ek + k(x + y) + d(x + y) + ed$

EXAMPLE 18

2 Factorise each expression.

a $p^2 - 81$

b $25 - h^2$

c $121 - a^2$

d $4r^2 - 9d^2$

e $25g^2 - 4e^2$

f $144 - 49m^2$

g $81y^2 - 16k^2$

h $1 - 4d^2$

i $m^2 - 25n^2$

j $m^2p^2 - 25$

k $a^2 - q^2w^2$

l $x^2 - \frac{1}{16}$

m $\frac{25}{9} - h^2$

n $6\frac{1}{4} - 9c^2$

o $16v^2 - \frac{9}{4}$

3 Factorise each expression.

a $4m^2 - 16p^2$

b $3y^2 - 27$

c $125a^2b^2 - 20v^2$

d $3x^4 - 27p^2$

e $54g^3p - 6gp^3$

f $28c - 63cw^2$

g $144e^5 - 36e^3$

h $9d^2 - 2\frac{1}{4}$

i $100k^2 - 6\frac{1}{4}$

j $98a^2 - 2w^2$

k $c^2 - 1\frac{7}{9}$

l $108m^3n^2 - 27m$

4 Factorise each expression.

a $\frac{1}{25} - x^2$

b $\frac{d^2}{9} - \frac{w^2}{49}$

c $3p^2 - \frac{48}{p^2}$

d $\frac{64b^2}{25} - \frac{4y^2}{81}$

e $u^4 - 16$

f $(m + 3)^2 - 25$

g $(x - y)^2 - y^2$

h $2a^5 - 162a$

i $(p + x)^2 - (p - x)^2$

j $\frac{4c^2}{9} - \frac{e^2}{36}$

k $4g^2 - (g - 2k)^2$

l $5w^5 - 80w$

Foundation Standard Complex

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Factorising quadratic expressions $x^2 + bx + c$

4.08

A **quadratic expression** such as $y^2 + 12y - 4$ is called a **trinomial** because it has 3 terms.

The expansion of $(x + 2)(x + 9)$ is $x^2 + 11x + 18$, a quadratic trinomial.

$\therefore$ The factorisation of $x^2 + 11x + 18$ is $(x + 2)(x + 9)$.

PATH

4.08

Video
Factorising quadratic expressions 1

Example 19

Factorise each quadratic trinomial.

a $a^2 + 7a + 12$ **b** $x^2 + 9x + 8$

Solution

a Find the 2 numbers that have a sum of 7 and a product of 12.

It is best to test numbers that have a **product of 12** and then check if their sums equal 7.

Pair of numbers	Product	Sum
6, 2	6 × 2 = 12	6 + 2 = 8
3, 4	3 × 4 = 12	3 + 4 = 7

The correct numbers are 3 and 4.

$\therefore a^2 + 7a + 12 = (a + 3)(a + 4)$

b Find 2 numbers with a sum of 9 and a product of 8.

Test numbers that have a **product of 8** and check if their sums equal 9.

The correct numbers are 8 and 1.

$\therefore x^2 + 9x + 8 = (x + 8)(x + 1)$

Factorising quadratic expressions of the form $x^2 + bx + c$

- Find 2 numbers that have a sum of b and a product of c
- Use these 2 numbers to write a binomial product of the form $(x ___)(x ___)$

Example 20

Factorise each quadratic expression.

a $x^2 + 2x - 24$ **b** $c^2 - c - 42$

Solution

a $x^2 + 2x - 24$

Find 2 numbers that have a product of -24 and a sum of 2.

Since the product is negative, one of the numbers must be negative.

They are $+6$ and -4.

$\therefore x^2 + 2x - 24 = (x + 6)(x - 4)$

b $c^2 - c - 42$

Product $= -42$, sum $= -1$.

Since the product is negative, one of the numbers must be negative.

They are -7 and $+6$.

$\therefore c^2 - c - 42 = (c - 7)(c + 6)$

PATH

EXAMPLES 19,20

EXERCISE 4.08 ANSWERS ON P. 620

Factorising quadratic expressions $x^2 + bx + c$ U F R

1 Factorise each quadratic expression. R

a	$k^2 + 13k + 36$	**b**	$a^2 + 19a + 90$	**c**	$g^2 + 10g + 21$
d	$m^2 - 11m + 28$	**e**	$p^2 - 8p + 15$	**f**	$c^2 - 15c + 54$
g	$u^2 - 5u + 4$	**h**	$h^2 - 14h + 48$	**i**	$e^2 - 8e + 16$
j	$d^2 + 3d - 28$	**k**	$q^2 + 2q - 80$	**l**	$n^2 + 4n - 45$
m	$x^2 - 6x - 7$	**n**	$g^2 - 11g - 80$	**o**	$w^2 - 4w - 96$
p	$r^2 - 12r - 28$	**q**	$b^2 - 3b - 54$	**r**	$m^2 - 4m - 12$
s	$y^2 - 14y + 49$	**t**	$d^2 - d - 30$	**u**	$e^2 + 9e - 90$

Quiz
Mental skills 4

Mental skills 4: Maths without calculators ANSWERS ON P. 620

Estimating answers

A quick way of estimating an answer is to round each number in the calculation.

1 Study each example.

a $55 + 132 - 34 + 17 - 78 \approx 60 + 130 - 30 + 20 - 80$
$= 100$ (Actual answer = 92)

b $510 \div 24 \approx 500 \div 20$
$= 50 \div 2$
$= 25$ (Actual answer = 21.25)

2 Now estimate each answer.

a	$27 + 11 + 87 + 142 + 64$	**b**	$55 + 34 - 22 - 46 + 136$
c	$684 + 903$	**d**	$35 + 81 + 110 + 22 + 7$
e	$517 - 96$	**f**	$210 - 38 - 71 + 151 - 49$
g	$766 - 353$	**h**	367×2
i	83×81	**j**	984×16
k	$828 \div 3$	**l**	$507 \div 7$

3 Study each example involving decimals.

a $4.78 \times 19.2 \approx 5 \times 20$
$= 100$ (Exact answer = 91.776)

b $\frac{37.6+9.3}{41.2-12.7} \approx \frac{38+9}{40-13}$
$= \frac{47}{27}$
$= \frac{50}{30}$
$= 1.6$ (Exact answer = 1.6456...)

Foundation Standard Complex

9780170479660

4 Now estimate each answer.

a $3.75 + 9.381 + 4.6 + 10.5$ **b** $14.807 + 6.6 - 7.22$ **c** 18.47×9.61

d 4.27×97.6 **e** $\frac{11.07+18.4}{12.2}$ **f** $\frac{38.18}{17.2-9.6}$

g $\frac{18.46 \times 4.9}{39.72-15.2}$ **h** $62.13 \div 10.7$ **i** $(4.89)^2$

Factorising quadratic expressions $ax^2 + bx + c$

4.09

We have factorised quadratic trinomials of the type $x^2 + bx + c$. For example, the factorisation of $x^2 + 6x + 8$ is $(x + 2)(x + 4)$. These are called **monic** quadratic expressions because x^2 does not have a **coefficient** (or its coefficient is 1).

We will now factorise quadratic trinomials of the type $ax^2 + bx + c$, such as $8x^2 + 14x - 15$, where x^2 has a coefficient. These are called **non-monic** quadratic expressions.

Example 21

Factorise $4y^2 + 8y - 140$

Solution

$4y^2 + 8y - 140 = 4(y^2 + 2y - 35)$ Taking out the HCF of 4 first.

$= 4(y + 7)(y - 5)$ Product $= -35$, sum $= 2$

Example 22

Factorise $5k^2 - 12k + 4$.

Solution

There is no HCF, so we need to split up the middle term $-12k$.

Find 2 numbers that have a product of 20 and a sum of -12.

$$5k^2 - 12k + 4 = 5k^2 - 12k + 4$$

(sum of -12 above; product of $5 \times 4 = 20$ below)

Since the sum is negative, at least one of the numbers must be negative.

Since the product is positive, **both** of the numbers must be negative.

The 2 numbers are -10 and -2, so we will split $-12k$ into $-10k$ and $-2k$.

$\therefore 5k^2 - 12k + 4 = 5k^2 - 10k - 2k + 4$

$= 5k(k - 2) - 2(k - 2)$ Factorising by grouping in pairs

$= (k - 2)(5k - 2)$ Factorising again

PATH

Puzzles
Product and factors squaresaw
Perfect squares
Factorominoes
Trinominoes
Factorising puzzle (Advanced)

Technology
Factorising trinomials

Spreadsheet
Factorising trinomials

Video
Factorising quadratic expressions 2

PATH

Factorising quadratic trinomials of the form $ax^2 + bx + c$

First, check whether you can factorise using the HCF; if not then follow the next steps

- Find 2 numbers that have a sum of b and a product of ac
- Use these 2 numbers to split the middle term bx into 2 terms
- Factorise by grouping in pairs



Videos
Factorising quadratic expressions (Advanced)

Advanced algebra

Example 23

Factorise each quadratic expression.

a $3x^2 - 11x + 10$ **b** $4x^2 - 3x - 7$

Solution

a $3x^2 - 11x + 10$

$3 \times 10 = 30$.

Find 2 numbers that have a product of 30 and a sum of -11.

Since the sum is negative and the product is positive, both numbers must be negative.

They are -6 and -5.

Split $-11x$ into $-6x$ and $-5x$.

$$\begin{aligned} 3x^2 - 11x + 10 &= 3x^2 - 6x - 5x + 10 \\ &= 3x(x - 2) - 5(x - 2) \\ &= (x - 2)(3x - 5) \end{aligned}$$

b $4x^2 - 3x - 7$

$4 \times (-7) = -28$.

Find 2 numbers with a product of -28 and a sum of -3.

Since the product is negative, one of the numbers must be negative. They are -7 and 4.

$$\begin{aligned} 4x^2 - 3x - 7 &= 4x^2 - 7x + 4x - 7 \\ &= x(4x - 7) + 1(4x - 7) \\ &= (4x - 7)(x + 1) \end{aligned}$$

Example 24

Factorise each quadratic expression.

a $24k^2 - 54k - 15$ **b** $14 + 29a - 15a^2$

Solution

a
$$\begin{aligned} 24k^2 - 54k - 15 &= 3(8k^2 - 18k - 5) && \text{Taking out the HCF of 3 first} \\ &= 3(8k^2 - 20k + 2k - 5) && \text{Product} = -40, \text{ sum} = -18 \\ &= 3[4k(2k - 5) + 1(2k - 5)] \\ &= 3(2k - 5)(4k + 1) \end{aligned}$$

b
$$\begin{aligned} 14 + 29a - 15a^2 &= -15a^2 + 29a + 14 && \text{Rearranging to make the } a^2 \text{ term first} \\ &= -(15a^2 - 29a - 14) && \text{Taking out a common factor of } -1 \\ &= -(15a^2 + 6a - 35a - 14) && \text{Product} = -210, \text{ sum} = -29 \\ &= -[3a(5a + 2) - 7(5a + 2)] \\ &= -(5a + 2)(3a - 7) && \text{This can also be written as } (5a + 2)(7 - 3a) \end{aligned}$$

EXERCISE 4.09 ANSWERS ON P. 620

Factorising quadratic expressions $ax^2 + bx + c$ UFR

PATH

4.09

1 Factorise each quadratic expression. Look for the highest common factor first. R

EXAMPLE 21

a $2k^2 + 16k + 30$ **b** $3q^2 - 3q - 90$ **c** $5x^2 - 55x + 140$

d $6y^2 + 12y - 18$ **e** $n^3 - 16n^2 + 63n$ **f** $24 + 2a - a^2$

g $4p + 32 - p^2$ **h** $-x^4 - 4x^3 + 21x^2$ **i** $-20 + 24h - 4h^2$

2 Factorise each quadratic expression. R

EXAMPLE 22

a $6d^2 + 19d + 15$ **b** $8m^2 + 10m + 3$ **c** $2y^2 + 7y + 5$

d $2w^2 + 31w + 15$ **e** $6k^2 + 23k + 20$ **f** $3u^2 + 22u + 35$

g $4x^2 + 21x + 27$ **h** $5e^2 + 39e + 28$ **i** $12h^2 + 13h + 3$

3 Factorise each quadratic expression. R

EXAMPLE 23

a $4k^2 - 11k + 6$ **b** $6w^2 - 17w + 5$ **c** $5p^2 - 23p + 12$

d $6g^2 - 35g + 49$ **e** $3x^2 - 32x + 20$ **f** $21m^2 - 26m + 8$

g $20r^2 - 23r + 6$ **h** $10a^2 - 7a - 6$ **i** $20u^2 - 23u - 21$

j $2p^2 - 17p - 9$ **k** $6y^2 - y - 40$ **l** $12b^2 - 17b - 5$

m $12q^2 - 8q - 15$ **n** $4h^2 + 13h - 12$ **o** $10n^2 + 31n - 14$

4 Factorise each expression given it is a perfect square. R

a $16w^2 - 56w + 49$ **b** $121h^2 + 176h + 64$ **c** $9p^2 - 90p + 225$

5 Factorise each quadratic expression. R

EXAMPLE 24

a $6m^2 - 9m - 60$ **b** $8x^3 - 18x^2 + 4x$ **c** $6 - 5e^2 - 29e$

d $-18w^2 - 51w + 42$ **e** $4 - 4k - 48k^2$ **f** $8 + 2n - 15n^2$

g $48d^2 + 156d + 90$ **h** $30h - 24h^2 + 9$ **i** $3 + 5p - 28p^2$

6 Factorise the quadratic expression $8a^2 + 14ax - 15x^2$. Select **A**, **B**, **C** or **D**. R

A $(4a - 3x)(2a + 5x)$ **B** $(2a - 5x)(4a + 3x)$

C $(5a + 4x)(3a - 2x)$ **D** $(4a - 5x)(2a + 3x)$

7 Factorise each quadratic expression. R

a $28d^2 - 11d + 1$ **b** $6 - 2y - 20y^2$ **c** $30w^2 + 41w - 15$

d $4p^2 + 23p + 15$ **e** $23a - 12 - 10a^2$ **f** $9r^2 + 29r + 20$

g $28k^2 + 19ky + 3y^2$ **h** $72p - 6p^2 - 3p^3$ **i** $4 - 21m - 18m^2$

j $10g - 54g^2 - 36g^3$ **k** $44u^2 - u - 3$ **l** $32h^2 + 28hm - 15m^2$

Foundation Standard Complex

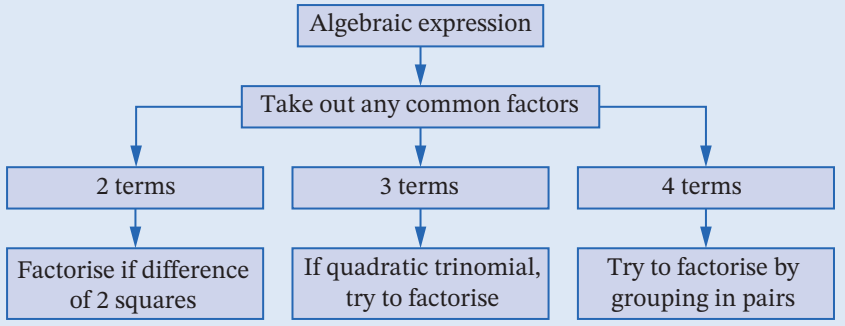
4.10 Mixed factorisations

PATH

Puzzles
Factorominoes
Mixed factorisations

Factorisation strategies

- Look for any common factors and factorise first
- If there are 2 terms, try factorising using the difference of 2 squares
- If there are 3 terms, try factorising as a quadratic trinomial
- If there are 4 terms, try factorising by grouping in pairs



Example 25

Factorise each quadratic expression.

a $4y^2 + 60d^2$ **b** $9m^3 - 18m^2 - 4m + 8$

Solution

a $4y^2 + 60d^2 = 4(y^2 + 15d^2)$ 2 terms but not a difference of 2 squares.

b $9m^3 - 18m^2 - 4m + 8 = 9m^2(m - 2) - 4(m - 2)$ Factorising by grouping in pairs

$= (m - 2)(9m^2 - 4)$

$= (m - 2)(3m + 2)(3m - 2)$ Difference of 2 squares

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EXERCISE 4.10 ANSWERS ON P. 620

Mixed factorisations U F R

1 Factorise each expression. R

a	$60y^2 - 15$	**b**	$15m^2 - 41m + 14$	**c**	$k^2 - 14k + 49$
d	$11a - 12 - 2a^2$	**e**	$6gy + 12cp + 8cy + 9gp$	**f**	$15 - 17x - 4x^2$
g	$64q^2 - 16q + 1$	**h**	$81r^2 - 36w^2$	**i**	$3bn^2 - 3np + 2bnd - 2dp$
j	$6k^2 + 23k + 20$	**k**	$42a^2 + 73a + 28$	**l**	$4 - 100h^2$
m	$36c^2 + 53c - 63$	**n**	$16 - 46d - 35d^2$	**o**	$25u^2 + 10u + 4$
p	$3xy - 8w + 4x - 6wy$	**q**	$1 - b^2 + b^3 - b$	**r**	$28c^2 - 9ch - 4h^2$
s	$15n^2 - 19n - 56$	**t**	$a^3p - ap$	**u**	$16r^2 - 146rk + 18k^2$

2 Factorise each expression. R

a	$48c^2 - 75$	**b**	$8q^2 + 22q - 40$	**c**	$15x^2 + x^3 - 2x^4$
d	$27p^2 - 24pr - 16r^2$	**e**	$9w^2 + 24w - 105$	**f**	$n^4 - 4n^2$
g	$y - y^2 - y^3 + y^4$	**h**	$36u^2 + 105u + 49$	**i**	$121k^2 - 198k + 81$
j	$4(c - d)^2 - (c + d)^2$	**k**	$20y^2 + 18d + 12y + 30dy$	**l**	$ah^2 - 6ah - ah^3$
m	$18b^2 - 54b + 27b - 81$	**n**	$32g - 8g^3y^2$	**o**	$8 - 42e - 36e^2$
p	$27q^2p - 3p$	**q**	$30h - 15h^2 - 15h^3$	**r**	$w^4 - w^3 - w^2 + 1$
s	$a^5 - 256a^3$	**t**	$m^2 - k^2 + 9m - 9k$	**u**	$30a^2 + 65a - 25$

3 Factorise the expression $w^6 - w^4 - w^2 + 1$. Select the correct answer **A**, **B**, **C** or **D**. R

A $w^4(w - 1)(w + 1)$ **B** $(w^2 - 1)(w^2 + 1)^2$

C $(w^4 - 1)^2$ **D** $(w^2 - 1)^2(w^2 + 1)$

PATH

EXAMPLE 25

4.11

Factorising algebraic fractions 4.11

Example 26

Simplify each expression.

a $\dfrac{30m-18}{25m^2-9}$ **b** $\dfrac{5k^2+9k-2}{k^2+7k+10}$

Solution

a $$\frac{30m-18}{25m^2-9} = \frac{6(5m-3)}{(5m+3)(5m-3)}$$
$$= \frac{6}{5m+3}$$

b $$\frac{5k^2+9k-2}{k^2+7k+10} = \frac{(5k-1)(k+2)}{(k+2)(k+5)}$$
$$= \frac{5k-1}{k+5}$$

PATH

Worksheet
Simplifying algebraic fractions

Foundation Standard Complex

PATH

Example 27

Simplify each expression.

a $\dfrac{5}{x^2+x-2}-\dfrac{2}{x^2-x}$ **b** $\dfrac{h^2+5h+6}{h^2+4h+3}\div\dfrac{h^2+2h}{h^2-2h-3}$

Solution

a $\dfrac{5}{x^2+x-2}-\dfrac{2}{x^2-x}=\dfrac{5}{(x+2)(x-1)}-\dfrac{2}{x(x-1)}$ Factorising denominators

$=\dfrac{5x}{x(x+2)(x-1)}-\dfrac{2(x+2)}{x(x+2)(x-1)}$ Using common denominators

$=\dfrac{5x-2(x+2)}{x(x+2)(x-1)}$

$=\dfrac{5x-2x-4}{x(x+2)(x-1)}$

$=\dfrac{3x-4}{x(x+2)(x-1)}$

b $\dfrac{h^2+5h+6}{h^2+4h+3}\div\dfrac{h^2+2h}{h^2-2h-3}=\dfrac{h^2+5h+6}{h^2+4h+3}\times\dfrac{h^2-2h-3}{h^2+2h}$

$=\dfrac{(h+2)(h+3)}{(h+1)(h+3)}\times\dfrac{(h-3)(h+1)}{h(h+2)}$

$=\dfrac{h-3}{h}$

EXERCISE 4.11 ANSWERS ON P. 620

Factorising algebraic fractions U F R

EXAMPLE 26

1 Simplify each expression. R

a $\dfrac{15x+9y}{3}$ **b** $\dfrac{6}{12r-18}$ **c** $\dfrac{4k-8d}{4}$

d $\dfrac{a^2-4}{a+2}$ **e** $\dfrac{3g+3h}{g^2-h^2}$ **f** $\dfrac{4p^2-16}{12p-24}$

g $\dfrac{w^2-49}{4(w+7)^2}$ **h** $\dfrac{7-y}{y-7}$ **i** $\dfrac{c^2+11c+30}{6c+30}$

j $\dfrac{he-hd+pe-pd}{e^2-d^2}$ **k** $\dfrac{k^2+5k-24}{2k^2-18}$ **l** $\dfrac{9y^2-4q^2}{3y^2-4yq-4q^2}$

m $\dfrac{r^2-2r-24}{r^2+12r+32}$ **n** $\dfrac{40+11b-2b^2}{8b^2+34b+35}$ **o** $\dfrac{6m-30-pm+5p}{m^2+3m-40}$

p $\dfrac{25-4a^2}{6a^2-11a-10}$ **q** $\dfrac{5n^2-4n-n^3}{n^2-2n+1}$ **r** $\dfrac{x^2y^2-4}{xy^2-3mxy-2y+6m}$

Foundation Standard Complex

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PATH

EXAMPLE 27

4.11

2 Simplify each expression. R

a $\frac{7}{k(k-1)}-\frac{4}{k(k+1)}$

b $\frac{8}{(x+4)(x-7)}+\frac{3}{x(x+4)}$

c $\frac{1}{(m+3)(2m+1)}+\frac{4}{(2m+1)(m-2)}$

d $\frac{5}{3h+24}-\frac{1}{h^2-64}$

e $\frac{10}{p^2+4p}+\frac{3}{p^2-4p}$

f $\frac{4}{a^2-25}-\frac{a}{(a+5)^2}$

g $\frac{2}{w^2-9w+20}+\frac{3}{w^2+2w-24}$

h $\frac{d}{d^2-3d-4}-\frac{7}{2d-8}$

i $\frac{9m}{16-25m^2}-\frac{5}{32m-40m^2}$

j $\frac{3u}{u^2+5u-14}-\frac{5}{2u^2-98}$

k $\frac{c+1}{10c^2+7c-12}+\frac{c}{5c^2-39c+28}$

l $\frac{n+1}{4n^3-36n}+\frac{2}{5n^2+15n}$

3 Simplify each expression. R

a $\frac{3r}{5r+25}\times\frac{r+5}{9}$

b $\frac{4y-12}{y}\times\frac{y^2+3y}{8y-24}$

c $\frac{w^2+6w}{2}\div\frac{w^2-36}{10w}$

d $\frac{8a-6}{3a^2+3a}\div\frac{12a-9}{a^3-a}$

e $\frac{5n}{n^2+10n+21}\div\frac{2n^2+4n}{n^2-2n-15}$

f $\frac{x^2+4x-5}{x^2+4x}\times\frac{x^2-16}{3x+15}$

g $\frac{15y-12k}{2y^2+ky-2y-k}\times\frac{8y+4k}{25y^2-16k^2}$

h $\frac{m^2-4m+4}{7m}\div\frac{5m-10}{21m^2+42m}$

i $\frac{1-9m^2}{m^2+9m+20}\times\frac{2m+10}{3m^2+5m-2}$

j $\frac{c^3+8c^2}{6c-18}\times\frac{c^2-6c+9}{c^2-64}$

k $\frac{3e}{e^2+5e-6}\div\frac{5e}{4-4e^2}$

l $\frac{2p-6}{2p^2-7p-4}\div\frac{p^2+3p-18}{p^2+3p-28}$

m $\frac{d^2+10d+21}{5d^2-20}\times\frac{d^2+4d-12}{d^2+6d-7}$

n $\frac{h^2+8h+12}{4h+12}\div\frac{h^2+10h+24}{h^2+4h+3}$

4.12 Simplifying surds

PATH



Puzzle
Simplifying surds

Technology
Simplifying surds quiz

Spreadsheet
Simplifying surds quiz

A **surd** is a square root ($\sqrt{\ }$), cube root ($\sqrt[3]{\ }$), or any type of root whose exact decimal or fraction value cannot be found. As a decimal, its digits run endlessly *without repeating* (like π), so they are neither terminating nor recurring decimals.

Rational numbers can be expressed in the form $\frac{a}{b}$, where a and b are integers ($b \neq 0$), but surds are **irrational numbers** because they cannot be expressed in this form. The rational and irrational numbers together make up the **real numbers**.

Rational numbers

can be expressed in the form $\frac{a}{b}$

Integers

$\frac{4}{1} = 4, \frac{26}{1} = 26, \frac{-3}{1} = -3$

Terminating decimals

$0.5, 7\frac{1}{8} = 7.125,$

$16\% = 0.16, 1.32$

Recurring decimals

$\frac{2}{3} = 0.666\ldots$

$\frac{5}{6} = 0.833\ldots$

$\frac{4}{11} = 0.3636\ldots$

Irrational numbers

cannot be expressed in the form $\frac{a}{b}$

Surds

$\sqrt{5}, -\sqrt{2}, \frac{\sqrt{11}}{3}, 8\sqrt{6}$

Transcendental numbers

Have no pattern and are non-recurring

e.g. $\pi = 3.14159...$,

$\cos 38° = 0.78801...$, $e = 2.71828...$

The square root of x

For $x < 0$ (negative), $\sqrt{x}$ is undefined.

For $x = 0$, $\sqrt{x} = 0$.

For $x > 0$ (positive), $\sqrt{x}$ is the positive square root of x.

For $x \geq 0$, $\left(\sqrt{x}\right)^2 = \sqrt{x} \times \sqrt{x} = x$ and $\sqrt{x^2} = x$.

The square root of a product

For $x > 0$ and $y > 0$:

$$\sqrt{xy} = \sqrt{x} \times \sqrt{y}$$

A surd $\sqrt{n}$ can be simplified if n can be divided into 2 factors, where one of them is a square number such as 4, 9, 16, 25, 36, 49, ...

Example 28

Simplify each surd.

a $\sqrt{108}$ **b** $4\sqrt{45}$ **c** $\dfrac{\sqrt{288}}{3}$

Solution

a *Method 1*:

$$\begin{aligned}\sqrt{108} &= \sqrt{36}\times\sqrt{3}\\ &= 6\times\sqrt{3}\\ &= 6\sqrt{3}\end{aligned}$$

Method 2:

$$\begin{aligned}\sqrt{108} &= \sqrt{4}\times\sqrt{27}\\ &= 2\times\sqrt{9}\times\sqrt{3}\\ &= 2\times 3\times\sqrt{3}\\ &= 6\sqrt{3}\end{aligned}$$

Method 2 involves simplifying surds *twice* ($\sqrt{108}$ and $\sqrt{27}$). Method 1 shows that when simplifying surds, look for the highest square factor possible.

b $$\begin{aligned}4\sqrt{45} &= 4\times\sqrt{9}\times\sqrt{5}\\ &= 4\times 3\times\sqrt{5}\\ &= 12\sqrt{5}\end{aligned}$$

c $$\begin{aligned}\frac{\sqrt{288}}{3} &= \frac{\sqrt{144}\times\sqrt{2}}{3}\\ &= \frac{12\sqrt{2}}{3}\\ &= \frac{12^{4}\sqrt{2}}{\cancel{3}^{1}}\\ &= 4\sqrt{2}\end{aligned}$$

EXERCISE 4.12 ANSWERS ON P. 621

Simplifying surds U F R

1 Simplify each expression.

a $\left(\sqrt{7}\right)^2$ **b** $\left(\sqrt{3}\right)^2$ **c** $\left(3\sqrt{2}\right)^2$ **d** $\left(4\sqrt{10}\right)^2$

e $\left(\sqrt{0.08}\right)^2$ **f** $\left(-2\sqrt{7}\right)^2$ **g** $\left(-3\sqrt{5}\right)^2$ **h** $\left(-5\sqrt{2}\right)^2$

2 Simplify each surd.

a $\sqrt{48}$ **b** $\sqrt{28}$ **c** $\sqrt{175}$ **d** $\sqrt{108}$

e $\sqrt{56}$ **f** $\sqrt{180}$ **g** $\sqrt{600}$ **h** $\sqrt{128}$

i $\sqrt{648}$ **j** $\sqrt{117}$ **k** $\sqrt{605}$ **l** $\sqrt{189}$

m $\sqrt{75}$ **n** $\sqrt{147}$ **o** $\sqrt{32}$ **p** $\sqrt{242}$

q $\sqrt{162}$ **r** $\sqrt{245}$ **s** $\sqrt{125}$ **t** $\sqrt{512}$

PATH

Videos
Simplifying surds 2
Simplifying surds 1

4.12

EXAMPLE 28

Foundation Standard Complex

PATH

3 Simplify each expression.

a $7\sqrt{20}$ **b** $2\sqrt{27}$ **c** $5\sqrt{80}$ **d** $\frac{\sqrt{98}}{4}$

e $\frac{\sqrt{120}}{6}$ **f** $5\sqrt{243}$ **g** $\frac{\sqrt{32}}{8}$ **h** $\frac{\sqrt{450}}{10}$

i $\frac{2}{5}\sqrt{125}$ **j** $\frac{\sqrt{3072}}{16}$ **k** $\frac{8}{3}\sqrt{243}$ **l** $4\sqrt{63}$

m $10\sqrt{160}$ **n** $3\sqrt{75}$ **o** $7\sqrt{68}$ **p** $\frac{\sqrt{52}}{6}$

4 Decide whether each statement is true (T) or false (F). R

a $2\sqrt{5} = \sqrt{10}$ **b** $\sqrt{24} = 12$ **c** $\left(\sqrt{5.7}\right)^2 = 5.7$

d $\sqrt{80} = 4\sqrt{5}$ **e** $\sqrt{11} \approx 3.3$ **f** The exact value of $\sqrt{12}$ is 3.464 101 6

Did you know?

Unreal numbers are imaginary!

$\sqrt{-2}$ is not a real number, because there is no real number which, if squared, equals -2. Numbers such as $\sqrt{-2}$, $\sqrt{-10}$, $\sqrt[4]{-17}$ are called **unreal** or **imaginary numbers** and cannot be graphed on a number line (that is, their values cannot be ordered).

Imaginary numbers were first noticed by the Greek mathematician Hero of Alexandria in the 1st century CE. Imaginary numbers were largely ignored until the 18th century when they were studied by Swiss mathematician Leonhard Euler and the German mathematician Carl Friedrich Gauss.

$\sqrt{-1}$ is defined to be the imaginary number i, so $\sqrt{-1} = i$ and $i^2 = -1$.

$\therefore \sqrt{-36} = \sqrt{36 \times (-1)} = \sqrt{36} \times \sqrt{-1} = 6i$.

Imaginary numbers are useful for solving physics and engineering problems involving heat conduction, elasticity, hydrodynamics and the flow of electric current.

Simplify each imaginary number:

a $\sqrt{-100}$ **b** $\sqrt{-25}$ **c** $\sqrt[4]{-16}$ **d** $\sqrt[6]{-64}$

Foundation Standard Complex

9780170479660

Adding and subtracting surds 4.13

Just as you can only add or subtract 'like terms' in algebra, you can only add or subtract 'like surds'. You may first need to express all the surds in their simplest forms.

PATH



4.13

Puzzle
Surds code puzzle

Example 29

Simplify each expression.

a $8\sqrt{6} - 5\sqrt{2} + 2\sqrt{6}$ **b** $\sqrt{48} + \sqrt{27}$

c $\sqrt{75} - \sqrt{32} + \sqrt{12}$ **d** $4\sqrt{50} - 2\sqrt{108}$

Solution

a $8\sqrt{6} - 5\sqrt{2} + 2\sqrt{6} = 10\sqrt{6} - 5\sqrt{2}$

b $\sqrt{48} + \sqrt{75} = \sqrt{16}\sqrt{3} + \sqrt{25}\sqrt{3}$
$= 4\sqrt{3} + 5\sqrt{3}$
$= 9\sqrt{3}$

c $\sqrt{27} - \sqrt{32} + \sqrt{12} = \sqrt{9}\sqrt{3} - \sqrt{16}\sqrt{2} + \sqrt{4}\sqrt{3}$
$= 3\sqrt{3} - 4\sqrt{2} + 2\sqrt{3}$
$= 5\sqrt{3} - 4\sqrt{2}$

d $4\sqrt{50} - 5\sqrt{72} = 4\sqrt{25}\sqrt{2} - 5\sqrt{36}\sqrt{2}$
$= 4 \times 5\sqrt{2} - 5 \times 6\sqrt{2}$
$= 20\sqrt{2} - 30\sqrt{2}$
$= -10\sqrt{2}$

EXERCISE 4.13 ANSWERS ON P. 621

Adding and subtracting surds U F

1 Simplify each expression.

a $5\sqrt{13} - 12 + 7\sqrt{13}$ **b** $12\sqrt{5} + 7\sqrt{3} + 8\sqrt{5}$

c $4\sqrt{2} - 5\sqrt{7} - 8\sqrt{2}$ **d** $4\sqrt{5} - \sqrt{11} - 7\sqrt{11} - 6\sqrt{5}$

e $9\sqrt{10} - 11\sqrt{15} - 2\sqrt{10} - 4\sqrt{15}$ **f** $2\sqrt{6} - 5 - 9\sqrt{6} + 4$

2 For each expression, select the correct simplified answer **A**, **B**, **C** or **D**.

EXAMPLE 29

a $2\sqrt{7} - 5\sqrt{28}$

A $4\sqrt{7}$ **B** $-3\sqrt{7}$ **C** $-12\sqrt{7}$ **D** $-8\sqrt{7}$

b $\sqrt{75} + \sqrt{300}$

A $\sqrt{375}$ **B** $15\sqrt{3}$ **C** $15\sqrt{6}$ **D** $12\sqrt{3}$

3 Simplify each expression.

a $\sqrt{32} - \sqrt{72}$ **b** $\sqrt{125} + \sqrt{45}$ **c** $\sqrt{44} + \sqrt{396}$ **d** $\sqrt{54} - \sqrt{216}$

e $2\sqrt{63} - 4\sqrt{28}$ **f** $7\sqrt{20} + \sqrt{45}$ **g** $\sqrt{300} + \sqrt{75}$ **h** $8\sqrt{18} - \sqrt{128}$

i $\sqrt{320} + 4\sqrt{125}$ **j** $\sqrt{98} - 5\sqrt{48}$ **k** $2\sqrt{44} + 6\sqrt{11}$ **l** $\sqrt{27} - \sqrt{48}$

m $\sqrt{150} - 2\sqrt{96} + \sqrt{54}$ **n** $5\sqrt{99} - 4\sqrt{45} - 7\sqrt{44}$ **o** $\sqrt{243} - 6\sqrt{50} - 10\sqrt{48}$

p $5\sqrt{24} + 3\sqrt{80} - 8\sqrt{96}$ **q** $2\sqrt{90} - 3\sqrt{200} - 2\sqrt{50}$ **r** $-\sqrt{68} + 3\sqrt{48} - 6\sqrt{27}$

Foundation Standard Complex

4.14 Multiplying and dividing surds

PATH

Puzzle
Surds

Worksheet
Multiplying and dividing surds

The square root of products and quotients

For $x > 0$ and $y > 0$:

$$\sqrt{xy} = \sqrt{x} \times \sqrt{y}$$

$$\sqrt{\frac{x}{y}} = \frac{\sqrt{x}}{\sqrt{y}}$$

Example 30

Simplify each expression.

a $3\sqrt{5} \times 4\sqrt{10}$ **b** $\sqrt{216} \div (-\sqrt{18})$ **c** $\frac{20\sqrt{150}}{5\sqrt{6}}$ **d** $\frac{8\sqrt{30} \times 5\sqrt{18}}{10\sqrt{90}}$

Solution

a
$$\begin{aligned} 3\sqrt{5} \times 4\sqrt{10} &= 3 \times 4 \times \sqrt{5} \times \sqrt{10} \\ &= 12\sqrt{50} \\ &= 12 \times \sqrt{25} \times \sqrt{2} \\ &= 12 \times 5\sqrt{2} \\ &= 60\sqrt{2} \end{aligned}$$

b
$$\begin{aligned} \sqrt{216} \div (-\sqrt{18}) &= -\frac{\sqrt{216}}{\sqrt{18}} \\ &= -\sqrt{12} \\ &= -\sqrt{4} \times \sqrt{3} \\ &= -\sqrt{3} \end{aligned}$$

c
$$\begin{aligned} \frac{20\sqrt{150}}{5\sqrt{6}} &= 4\sqrt{25} \\ &= 4 \times 5 \\ &= 20 \end{aligned}$$

d
$$\begin{aligned} \frac{8\sqrt{30} \times 5\sqrt{18}}{10\sqrt{90}} &= \frac{40\sqrt{540}}{10\sqrt{90}} \\ &= 4\sqrt{6} \end{aligned}$$

EXERCISE 4.14 ANSWERS ON P. 621

Multiplying and dividing surds U F

EXAMPLE 30

1 Simplify each expression.

a $5\sqrt{7} \times (-3\sqrt{3})$ **b** $6\sqrt{10} \times 4\sqrt{2}$ **c** $4\sqrt{3} \times (-2\sqrt{6})$

d $7\sqrt{8} \times 2\sqrt{5}$ **e** $8\sqrt{2} \times 3\sqrt{6}$ **f** $3\sqrt{6} \times 9\sqrt{24}$

g $-4\sqrt{20} \times 3\sqrt{15}$ **h** $5\sqrt{54} \times 4\sqrt{18}$ **i** $-6\sqrt{27} \times 4\sqrt{60}$

j $2\sqrt{300} \times 4\sqrt{15}$ **k** $3\sqrt{44} \times (-2\sqrt{99})$ **l** $5\sqrt{8} \times 4\sqrt{40}$

m $8\sqrt{3} \times 3\sqrt{54}$ **n** $-8\sqrt{32} \times \sqrt{27}$ **o** $\sqrt{90} \times \sqrt{72}$

p $-5\sqrt{20} \times 3\sqrt{8}$ **q** $7\sqrt{18} \times 3\sqrt{24}$ **r** $3\sqrt{48} \times 2\sqrt{12}$

Foundation Standard Complex

9780170479660

2 Simplify each expression.

a $-\sqrt{35} \div \sqrt{7}$ **b** $\sqrt{30} \div \sqrt{10}$ **c** $\frac{\sqrt{84}}{\sqrt{6}}$

d $8\sqrt{24} \div 2\sqrt{6}$ **e** $-15\sqrt{48} \div 4\sqrt{6}$ **f** $6\sqrt{125} \div 10\sqrt{5}$

g $8\sqrt{8} \div 5\sqrt{32}$ **h** $\frac{\sqrt{512}}{5\sqrt{8}}$ **i** $\frac{3\sqrt{72}}{2\sqrt{18}}$

j $5\sqrt{80} \div \left(-2\sqrt{10}\right)$ **k** $-42\sqrt{24} \div 3\sqrt{98}$ **l** $2\sqrt{15} \div 3\sqrt{90}$

m $\frac{-4\sqrt{24}}{6\sqrt{96}}$ **n** $20\sqrt{45} \div 5\sqrt{15}$ **o** $6\sqrt{56} \div 2\sqrt{14}$

p $12\sqrt{63} \div 3\sqrt{7}$ **q** $\frac{8\sqrt{50}}{2\sqrt{200}}$ **r** $6\sqrt{3} \div \sqrt{243}$

3 Simplify:

a $6\sqrt{5} \times 3\sqrt{5}$ **b** $8\sqrt{y} \times 4\sqrt{y}$ **c** $\sqrt{k^2} \times 2\sqrt{k}$

4 Simplify each expression.

a $\frac{8\sqrt{60} \times 5\sqrt{15}}{2\sqrt{20}}$ **b** $\frac{4\sqrt{24} \times 6\sqrt{18}}{8\sqrt{75}}$ **c** $\frac{2\sqrt{21} \times \sqrt{14}}{7\sqrt{98}}$

d $\frac{9\sqrt{15}}{2\sqrt{10} \times 4\sqrt{24}}$ **e** $\frac{2\sqrt{405} \times 8\sqrt{15}}{3\sqrt{75} \times 6\sqrt{12}}$ **f** $\frac{8\sqrt{80} \times 3\sqrt{2}}{4\sqrt{5} \times 6\sqrt{8}}$

PATH

4.15

Binomial products involving surds 4.15

Example 31

Expand and simplify each expression.

a $\sqrt{3}\left(\sqrt{5}+\sqrt{7}\right)$ **b** $2\sqrt{11}\left(3\sqrt{11}-5\sqrt{2}\right)$

Solution

a $\sqrt{3}\left(\sqrt{5}+\sqrt{7}\right) = \sqrt{3} \times \sqrt{5} + \sqrt{3} \times \sqrt{7}$

$= \sqrt{15} + \sqrt{21}$

b $2\sqrt{11}\left(3\sqrt{11}-5\sqrt{2}\right) = 2\sqrt{11} \times 3\sqrt{11} - 2\sqrt{11} \times 5\sqrt{2}$

$= 6 \times 11 - 10 \times \sqrt{22}$

$= 66 - 10\sqrt{22}$

PATH

Foundation Standard Complex

PATH

Example 32

Expand and simplify each binomial product.

a $(\sqrt{7}+\sqrt{5})(3\sqrt{2}-\sqrt{3})$ **b** $(3-2\sqrt{10})(\sqrt{5}-3\sqrt{2})$

Solution

a $(\sqrt{7}+\sqrt{5})(3\sqrt{2}-\sqrt{3}) = \sqrt{7}(3\sqrt{2}-\sqrt{3})+\sqrt{5}(3\sqrt{2}-\sqrt{3})$

$= \sqrt{7}\times 3\sqrt{2}-\sqrt{7}\times\sqrt{3}+\sqrt{5}\times 3\sqrt{2}-\sqrt{5}\times\sqrt{3}$

$= 3\sqrt{14}-\sqrt{21}+3\sqrt{10}-\sqrt{15}$

b $(3-2\sqrt{10})(\sqrt{5}-3\sqrt{2}) = 3(\sqrt{5}-3\sqrt{2})-2\sqrt{10}(\sqrt{5}-3\sqrt{2})$

$= 3\sqrt{5}-9\sqrt{2}-2\sqrt{50}+6\sqrt{20}$

$= 3\sqrt{5}-9\sqrt{2}-2(5\sqrt{2})+6(2\sqrt{5})$

$= 3\sqrt{5}-9\sqrt{2}-10\sqrt{2}+12\sqrt{5}$

$= 15\sqrt{5}-19\sqrt{2}$

Example 33

Video
Binomial products involving surds

Expand and simplify each expression.

a $(\sqrt{7}-\sqrt{5})^2$ **b** $(2\sqrt{3}+3\sqrt{5})^2$

c $(\sqrt{5}-\sqrt{2})(\sqrt{5}+\sqrt{2})$ **d** $(3\sqrt{11}+4)(3\sqrt{11}-4)$

Solution

a $(\sqrt{7}-\sqrt{5})^2 = (\sqrt{7})^2-2\times\sqrt{7}\times\sqrt{5}+(\sqrt{5})^2$ using $(a-b)^2 = a^2-2ab+b^2$

$= 7-2\sqrt{35}+5$

$= 12-2\sqrt{35}$

b $(2\sqrt{3}+3\sqrt{5})^2 = (2\sqrt{3})^2+2\times 2\sqrt{3}\times 3\sqrt{5}+(3\sqrt{5})^2$ using $(a+b)^2 = a^2+2ab+b^2$

$= (4\times 3)+12\sqrt{15}+(9\times 5)$

$= 12+12\sqrt{15}+45$

$= 57+12\sqrt{15}$

c $(\sqrt{5}-\sqrt{2})(\sqrt{5}+\sqrt{2}) = (\sqrt{5})^2-(\sqrt{2})^2$ using $(a+b)(a-b) = a^2-b^2$

$= 5-2$

$= 3$

Note that because of the 'difference of 2 squares', the answer is not a surd but a rational number.

d $(3\sqrt{11}+4)(3\sqrt{11}-4) = (3\sqrt{11})^2-4^2$ using $(a+b)(a-b) = a^2-b^2$

$= (9\times 11)-16$

$= 83$

EXERCISE 4.15 ANSWERS ON P. 621

PATH

Binomial products involving surds U F

EXAMPLE 31

1 Expand and simplify each expression.

a $\sqrt{5}\left(\sqrt{3}+\sqrt{2}\right)$ **b** $\sqrt{6}\left(\sqrt{2}-1\right)$ **c** $\sqrt{2}\left(\sqrt{3}+\sqrt{7}\right)$

d $\sqrt{5}\left(3\sqrt{2}-\sqrt{5}\right)$ **e** $3\sqrt{2}\left(\sqrt{2}+2\sqrt{3}\right)$ **f** $-\sqrt{11}\left(4-\sqrt{5}\right)$

g $2\sqrt{7}\left(3\sqrt{7}-4\right)$ **h** $5\sqrt{5}\left(1+3\sqrt{5}\right)$ **i** $3\sqrt{2}\left(4\sqrt{2}+\sqrt{3}\right)$

4.15

EXAMPLE 32

2 Expand and simplify $\left(\sqrt{3}+2\sqrt{5}\right)\left(5\sqrt{2}+\sqrt{3}\right)$. Select the correct answer **A**, **B**, **C** or **D**.

A $20\sqrt{10}$ **B** $2\sqrt{15}+5\sqrt{6}$

C $5\sqrt{6}+3+10\sqrt{10}+2\sqrt{15}$ **D** $5\sqrt{5}+\sqrt{3}+7\sqrt{7}+4\sqrt{2}$

3 Expand and simplify each expression.

a $\left(\sqrt{5}-3\right)\left(2\sqrt{5}+\sqrt{2}\right)$ **b** $\left(\sqrt{7}-\sqrt{3}\right)\left(\sqrt{7}+2\right)$

c $\left(7\sqrt{3}+2\right)\left(4\sqrt{2}+\sqrt{3}\right)$ **d** $\left(3\sqrt{2}-\sqrt{5}\right)\left(5\sqrt{2}+2\sqrt{5}\right)$

e $\left(\sqrt{7}+2\sqrt{11}\right)\left(3\sqrt{7}+4\sqrt{11}\right)$ **f** $\left(5\sqrt{3}-2\sqrt{2}\right)\left(4\sqrt{3}-3\sqrt{2}\right)$

g $\left(6+2\sqrt{10}\right)\left(3\sqrt{10}-1\right)$ **h** $\left(\sqrt{7}-2\sqrt{5}\right)\left(3\sqrt{5}+2\sqrt{7}\right)$

EXAMPLE 33

4 Expand $\left(5+\sqrt{7}\right)^2$. Select **A**, **B**, **C** or **D**.

A 12 **B** 32 **C** $32+10\sqrt{7}$ **D** $32+5\sqrt{7}$

5 Expand and simplify each expression.

a $\left(\sqrt{5}-\sqrt{3}\right)^2$ **b** $\left(\sqrt{7}+\sqrt{2}\right)^2$ **c** $\left(\sqrt{5}-2\right)^2$

d $\left(3+\sqrt{10}\right)^2$ **e** $\left(5\sqrt{2}+3\sqrt{3}\right)^2$ **f** $\left(5\sqrt{7}-2\right)^2$

g $\left(3\sqrt{2}+2\sqrt{5}\right)^2$ **h** $\left(2\sqrt{5}+\sqrt{3}\right)^2$

6 Expand and simplify each expression.

a $\left(\sqrt{3}-\sqrt{2}\right)\left(\sqrt{3}+\sqrt{2}\right)$ **b** $\left(5+\sqrt{3}\right)\left(5-\sqrt{3}\right)$ **c** $\left(6+2\sqrt{7}\right)\left(6-2\sqrt{7}\right)$

d $\left(\sqrt{5}-\sqrt{3}\right)\left(\sqrt{5}+\sqrt{3}\right)$ **e** $\left(\sqrt{11}-\sqrt{10}\right)\left(\sqrt{11}+\sqrt{10}\right)$ **f** $\left(5\sqrt{7}+3\right)\left(5\sqrt{7}-3\right)$

g $\left(3\sqrt{2}+\sqrt{5}\right)\left(3\sqrt{2}-\sqrt{5}\right)$ **h** $\left(4\sqrt{2}-5\sqrt{3}\right)\left(4\sqrt{2}+5\sqrt{3}\right)$

7 Expand and simplify $\left(5\sqrt{2}-4\sqrt{3}\right)\left(5\sqrt{2}+4\sqrt{3}\right)$. Select **A**, **B**, **C** or **D**.

A $25\sqrt{2}-16\sqrt{3}$ **B** $10\sqrt{2}+10\sqrt{6}$ **C** 2 **D** 26

8 Expand and simplify each expression.

a $\left(3\sqrt{7}-5\right)^2$ **b** $\left(5\sqrt{2}-4\right)\left(\sqrt{2}+5\right)$ **c** $\left(2\sqrt{7}+3\sqrt{5}\right)\left(\sqrt{5}+\sqrt{7}\right)$

d $\left(4\sqrt{3}+5\right)^2$ **e** $\left(4\sqrt{2}+\sqrt{3}\right)\left(4\sqrt{2}-\sqrt{3}\right)$ **f** $\left(3\sqrt{10}-\sqrt{2}\right)^2$

Foundation Standard Complex

PATH

Investigation

Making the denominator rational

1 What happens when we multiply the numerator and denominator of a fraction by the same number?

2 **a** Simplify $\frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$.

b Mentally approximate the value of $\frac{\sqrt{2}}{2}$ given $\sqrt{2} \approx 1.4142$.

c Check using a calculator that $\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$. Why is this true?

3 **a** Is it true that $\frac{3}{\sqrt{7}} = \frac{3}{\sqrt{7}} \times \frac{\sqrt{7}}{\sqrt{7}}$? Why?

b Simplify $\frac{3}{\sqrt{7}} \times \frac{\sqrt{7}}{\sqrt{7}}$. Compare your answer with those of other students.

c Check using a calculator that $\frac{3}{\sqrt{7}} = \frac{3\sqrt{7}}{7}$.

4 **a** Explain why $\frac{3\sqrt{5}}{\sqrt{2}} = \frac{3\sqrt{5}}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$.

b Show that $\frac{3\sqrt{5}}{\sqrt{2}} = \frac{3\sqrt{10}}{2}$.

4.16 Rationalising the denominator

PATH

If $\sqrt{2} \approx 1.4142$, what is the value of $\frac{3}{\sqrt{2}}$? Fractions containing surds in the denominator are difficult to work with. When approximating the value of $\frac{3}{\sqrt{2}}$, it is difficult to mentally divide by 1.4142. We can overcome this by making the denominator **rational** (that is, **not** a surd). Numbers that have irrational denominators, such as $\frac{1}{\sqrt{5}}, \frac{3}{2\sqrt{7}}, \frac{\sqrt{3}}{\sqrt{2}}, \frac{5\sqrt{7}}{\sqrt{3}}$ can be rewritten with a rational denominator by multiplying both the numerator and denominator by the surd that appears in the denominator. This method is called **rationalising the denominator**.

Video
Rationalising the denominator

Example 34

Rationalise the denominator of each surd.

a $\frac{3}{\sqrt{2}}$ **b** $\frac{5}{4\sqrt{3}}$ **c** $\frac{8\sqrt{2}}{3\sqrt{5}}$ **d** $\frac{\sqrt{2}+1}{\sqrt{3}}$

Solution

a $\frac{3}{\sqrt{2}} = \frac{3}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$ because $\frac{\sqrt{2}}{\sqrt{2}} = 1$

$= \frac{3\sqrt{2}}{2}$

Note: Now it is easier to approximate $\frac{3}{\sqrt{2}}$ by approximating $\frac{3\sqrt{2}}{2}$ and mentally multiplying $\frac{3}{2}$ by 1.4142 than by dividing 3 by 1.4142.

b $\frac{5}{4\sqrt{3}} = \frac{5}{4\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$

$= \frac{5\sqrt{3}}{4 \times 3}$

$= \frac{5\sqrt{3}}{12}$

c $\frac{8\sqrt{2}}{3\sqrt{5}} = \frac{8\sqrt{2}}{3\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}}$

$= \frac{8\sqrt{10}}{3 \times 5}$

$= \frac{8\sqrt{10}}{15}$

d $\frac{\sqrt{2}+1}{\sqrt{3}} = \frac{\sqrt{2}+1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$

$= \frac{\sqrt{6}+\sqrt{3}}{3}$

PATH

4.16

EXERCISE 4.16 ANSWERS ON P. 621

Rationalising the denominator U F

EXAMPLE 34

1 By rationalising the denominator, which surd is equivalent to $\frac{2}{\sqrt{6}}$?
Select the correct answer **A**, **B**, **C** or **D**.

A $2\sqrt{6}$ **B** $\frac{\sqrt{6}}{3}$ **C** $\frac{\sqrt{6}}{6}$ **D** $\frac{2\sqrt{6}}{3}$

2 Rationalise the denominator of each surd.

a $\frac{1}{\sqrt{2}}$ **b** $\frac{1}{\sqrt{7}}$ **c** $\frac{1}{\sqrt{3}}$ **d** $\frac{3}{\sqrt{2}}$ **e** $\frac{2}{\sqrt{7}}$ **f** $\frac{1}{3\sqrt{2}}$

g $\frac{1}{2\sqrt{3}}$ **h** $\frac{1}{4\sqrt{7}}$ **i** $\frac{7}{3\sqrt{5}}$ **j** $\frac{\sqrt{2}}{3\sqrt{5}}$ **k** $\frac{3\sqrt{2}}{2\sqrt{6}}$ **l** $\frac{5\sqrt{3}}{4\sqrt{5}}$

3 Which surd is equivalent to $\frac{\sqrt{3}}{2\sqrt{5}}$? Select **A**, **B**, **C** or **D**.

A $\frac{\sqrt{15}}{10}$ **B** $2\sqrt{15}$ **C** $\frac{\sqrt{15}}{3}$ **D** $\sqrt{5}$

4 Which surd is equivalent to $\frac{\sqrt{27}}{3\sqrt{18}}$? Select **A**, **B**, **C** or **D**.

A $\frac{1}{2}$ **B** $\frac{\sqrt{2}}{2}$ **C** $\frac{\sqrt{5}}{6}$ **D** $\frac{\sqrt{6}}{6}$

5 Rationalise the denominator of each expression.

a $\frac{\sqrt{2}-1}{\sqrt{2}}$ **b** $\frac{1-\sqrt{5}}{\sqrt{5}}$ **c** $\frac{5+\sqrt{3}}{2\sqrt{2}}$ **d** $\frac{\sqrt{2}-\sqrt{3}}{3\sqrt{6}}$

6 Simplify each expression, giving the answer with a rational denominator.

a $\frac{1}{\sqrt{7}} + \frac{1}{\sqrt{2}}$ **b** $\frac{\sqrt{2}}{\sqrt{5}} + \frac{3}{\sqrt{3}}$ **c** $\frac{3}{2\sqrt{3}} - \frac{1}{\sqrt{2}}$

Foundation Standard Complex

Power plus ANSWERS ON P. 622

1 Expand and simplify each expression.

a $(x+5)^3$ **b** $(y-2)^3$ **c** $(a+b)^3$ **d** $(3d+10)^3$

2 Use the given expansion to evaluate each square number without using a calculator.

a $21^2 = (20+1)^2$ **b** $45^2 = (40+5)^2$

c $29^2 = (30-1)^2$ **d** $59^2 = (60-1)^2$

e $102^2 = (100+2)^2$ **f** $98^2 = (100-2)^2$

3 By expressing 31×29 as $(30+1)(30-1)$, evaluate 31×29 without using a calculator.

4 Use the method of question **3** to evaluate each expression.

a 21×19 **b** 51×49 **c** 89×91 **d** 78×82

5 **a** Is it true that $\frac{1}{3+\sqrt{2}} = \frac{1}{3+\sqrt{2}} \times \frac{3-\sqrt{2}}{3-\sqrt{2}}$? Explain.

b Simplify $\frac{1}{3+\sqrt{2}} \times \frac{3-\sqrt{2}}{3-\sqrt{2}}$. Is the denominator rational?

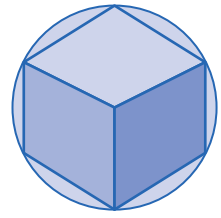
c Use a calculator to check that the value of your answer to part **b** is equal to the value of $\frac{1}{3+\sqrt{2}}$.

6 The **conjugate** of $3+\sqrt{2}$ is $3-\sqrt{2}$. The process shown in question **5** involves rationalising a surd with a **binomial** denominator. By first finding the **conjugate** of the denominator, rationalise the denominator of each expression below.

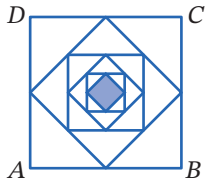
a $\frac{1}{2-\sqrt{3}}$ **b** $\frac{\sqrt{2}}{\sqrt{5}+1}$

c $\frac{1}{\sqrt{7}-\sqrt{3}}$ **d** $\frac{2}{5+2\sqrt{3}}$

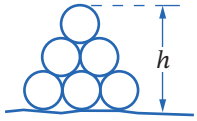
7 The largest cube that can fit inside a sphere must have its 8 vertices touching the surface of the sphere. Express the side length, s, of the cube in terms of the diameter, D, of the sphere.



8 Squares are formed inside squares by joining the midpoints of the sides of the squares as shown. If $AB = 4$ cm, find the exact length of the side of the shaded square.



9 6 stormwater pipes, each 2 mm in diameter, are stacked as shown in the diagram. Find the exact height, h, of the stack.



CHAPTER 4 REVIEW

Quiz
Language of maths 4

Worksheet
Algebra 6

Puzzles
Algebra crossword

Surds crossword

Language of maths

algebraic fraction	binomial product	coefficient	constant term
denominator	difference of 2 squares	factorise	fractional
grouping in pairs	index law	indices	irrational
perfect square	product	quadratic expression	quotient
rational	rationalise	real	reciprocal
root	square number	surd	trinomial

1 Explain the meaning of **quadratic** and **trinomial**. Give an example of a **quadratic trinomial**.

2 What power is associated with the **reciprocal** of a term or number?

3 Copy and complete:

To factorise quadratic expressions of the form $x^2 + bx + c$, first find 2 numbers that have a _____ of b and a _____ of c.

4 How do you **simplify** a surd?

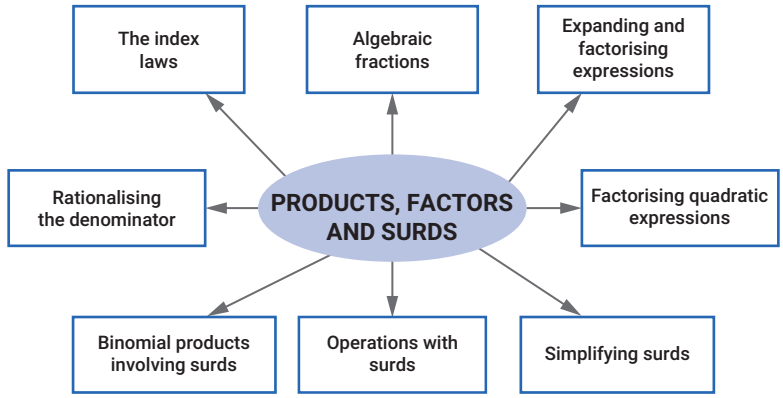
5 Why is π an example of an **irrational number** that is not a **surd**?

6 How do you **rationalise the denominator** of a surd expression?

Worksheet
Mind map: Products, factors and surds

Topic summary

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



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TEST YOURSELF 4

ANSWERS ON P. 622

Quiz
Test yourself 4

4.01

PATH

1 Simplify each expression.

a $3v^4w^3 \times 2v^5w^8$ **b** $\dfrac{24r^8h^8}{3rh^2}$ **c** $(5xy^2)^2$

d $\left(\dfrac{2p}{3}\right)^0$ **e** $(4k)^{-1}$ **f** $\left(\dfrac{2p}{3}\right)^3$

g $(4m)^{-2}$ **h** $4m^{-2}$ **i** $\left(\dfrac{5b^8y^6}{b^2y^3}\right)^4$

j $(4t^4u^5)^3 \times 8t^2u$ **k** $45c^6d^8 \div (-3cd^2)^2$ **l** $\left(\dfrac{45ab^4}{54a^2b^3}\right)^{-1}$

4.01

2 Simplify each expression.

a $\left(\dfrac{1}{10}\right)^{-5}$ **b** $\left(\dfrac{3}{5}\right)^{-2}$ **c** $\left(\dfrac{4g}{7}\right)^{-3}$ **d** $\left(\dfrac{9}{2d}\right)^{-2}$

4.02

3 Evaluate each expression.

a $400^{\frac{1}{2}}$ **b** $(-27)^{\frac{5}{3}}$ **c** $64^{-\frac{1}{2}}$ **d** $(-32)^{\frac{2}{5}}$

4.02

4 Evaluate each expression.

a $\left(16a^4\right)^{\frac{3}{2}}$ **b** $\sqrt[4]{\left(81m^{12}\right)^5}$ **c** $\dfrac{1}{\left(\sqrt[3]{8p^6}\right)}$ **d** $\left(125x^3y^6\right)^{-\frac{2}{3}}$

4.03

5 Simplify each expression.

a $\dfrac{r}{4} - \dfrac{3r}{5}$ **b** $\dfrac{5g}{3} + \dfrac{3g}{2}$ **c** $\dfrac{11x}{24} - \dfrac{2x}{3}$ **d** $\dfrac{3}{4} - \dfrac{2m}{7}$

PATH

4.03

6 Simplify each expression.

a $\dfrac{3k-4}{5} + \dfrac{2k}{3}$ **b** $\dfrac{w+1}{4} - \dfrac{w-3}{9}$ **c** $\dfrac{2p+5}{3} - \dfrac{7p}{12}$ **d** $\dfrac{5+2y}{11} + \dfrac{1-y}{8}$

4.04

7 Simplify each expression.

a $\dfrac{4m}{5} \times \dfrac{3m}{8}$ **b** $\dfrac{6}{d} \times \dfrac{4de}{21}$ **c** $\dfrac{5}{a} \div \dfrac{10}{b}$ **d** $\dfrac{vw^3}{5} \div \dfrac{3v}{20} \times \dfrac{4}{3w}$

4.05

8 Expand and simplify each expression.

a $8pq(7p - 5q)$ **b** $-5h^2(3h + 7)$ **c** $4eg(g - 6e) - 6e^2g$

d $12(9 - n) - 5(2n + 3)$ **e** $x^2(6x + x^2) + 2x(5x^3 + x^2)$ **f** $3(7 - 2y) - 5y(3 + 4y)$

4.05

9 Factorise each expression.

a $16ar^2 + 24ar$ **b** $-24p + 18q$ **c** $2(5x - 1) - 3x(5x - 1)$

d $15xy^2 - 30x^3y^3$ **e** $6pt^2 + 12p^2t - 48p^3$ **f** $32r^2m^4 + 12r^4m^3$

g $50x^4y^3 - 75x^3y^4$ **h** $-8p^3q^3 + 48p^3q^6$ **i** $n(n^2 + 6) - (n^2 + 6)$

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4.06

10 Expand each binomial product.

a $(b + 3)(b + 10)$ **b** $(d + 8)(d - 10)$ **c** $(w - 6)(9 - w)$

d $(5x + 7)(4x - 3)$ **e** $(7y - 3)(7y + 3)$ **f** $(5p - 8)(5p - 8)$

PATH

4.07

11 Expand each binomial product.

a $(n + 9)(n - 9)$ **b** $(3y + 2d)^2$ **c** $(4n - 11)(4n + 11)$

4.07

12 Factorise each expression.

a $3pu + 2qt + 2qu + 3pt$ **b** $4ab + 6bc - 6ad - 9dc$ **c** $b^2 - 100$

d $25 - 16y^2$ **e** $20x^2 - 125$ **f** $3r^3 - 27r$

4.08

13 Factorise each expression.

a $y^2 + 10y + 25$ **b** $q^2 - 21q + 20$ **c** $n^2 + 8n - 33$

d $a^2 - 11a + 28$ **e** $m^2 - 5m - 84$ **f** $p^2 + 3p - 54$

4.09

14 Factorise each expression.

a $3w^2 + 5w + 2$ **b** $2y^2 - 3y - 9$ **c** $60 - 5b - 5b^2$

d $3p^2 + 10p - 8$ **e** $12x^2 - 46x + 14$ **f** $6n^2 - 13n + 6$

g $40y^2 + 49y - 24$ **h** $48c^2 - 64c - 35$ **i** $10e^2 - 7e + 1$

4.10

15 Factorise each expression.

a $5q^2 - 45$ **b** $20x^2 - 52x + 24$ **c** $c^3 - c^2 - c + 1$

d $4 - 47k - 12k^2$ **e** $6m^2 + m - 40$ **f** $a^4 - a^2 + a^3 - a$

4.11

16 Simplify each expression.

a $\frac{20d - 35}{10}$ **b** $\frac{4r + 12y}{7r^2 + 21ry}$

c $\frac{u^2 - 4u - 32}{2u^2 - 128}$ **d** $\frac{1}{25 - h^2} - \frac{3}{h^2 + 13h + 40}$

e $\frac{4}{p^2 - 9} + \frac{2}{p^2 + 6p + 9}$ **f** $\frac{6m^2 - 24}{7m + 14}$

g $\frac{3}{y^2 + 4y} - \frac{y}{y^2 - 16}$ **h** $\frac{3w - 15}{5} \times \frac{20w}{3w^2 - 15w}$

i $\frac{b^2 + 3b + 2}{b^2 - 9} \div \frac{b^2 + b}{3b + 9}$ **j** $\frac{6a}{a^2 + 5a + 6} - \frac{2}{a^2 - 5a - 14}$

k $\frac{3 - 12q^2}{q^2 + 4q + 4} \times \frac{q^2 + q - 2}{6q - 12q^2}$ **l** $\frac{r^3 - 25r}{r^2 - 9} \div \frac{r^2 + 12r + 35}{12r - 36}$

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17 Simplify each surd. 4.12

a $\sqrt{162}$ **b** $\sqrt{48}$ **c** $\sqrt{175}$

d $\sqrt{512}$ **e** $4\sqrt{250}$ **f** $3\sqrt{56}$

g $5\sqrt{288}$ **h** $8\sqrt{45}$ **i** $2\sqrt{28}$

j $\frac{1}{2}\sqrt{24}$ **k** $\frac{2}{3}\sqrt{3125}$ **l** $\frac{1}{2}\sqrt{32}$

m $2\sqrt{44}$ **n** $\frac{2}{5}\sqrt{225}$ **o** $\frac{2}{3}\sqrt{162}$

18 Simplify each expression. 4.13

a $\sqrt{600}+\sqrt{24}$ **b** $13\sqrt{5}+\sqrt{50}-4\sqrt{80}$

c $6\sqrt{32}-\sqrt{27}-3\sqrt{98}+4\sqrt{75}$ **d** $6\sqrt{45}-3\sqrt{63}+5\sqrt{80}$

e $\sqrt{800}-2\sqrt{243}+3\sqrt{72}-2\sqrt{27}$ **f** $7\sqrt{44}-2\sqrt{99}$

19 Simplify each expression. 4.14

a $\sqrt{10}\times\sqrt{7}$ **b** $\sqrt{8}\times\sqrt{12}$ **c** $\sqrt{6}\times\sqrt{18}$

d $\sqrt{5}\times\sqrt{10}$ **e** $\sqrt{80}\div\sqrt{8}$ **f** $\sqrt{98}\div\sqrt{2}$

g $8\sqrt{42}\div 2\sqrt{6}$ **h** $\sqrt{350}\div 5\sqrt{7}$ **i** $\frac{\sqrt{75}}{3\sqrt{3}}$

j $\frac{\sqrt{18}\times\sqrt{30}}{\sqrt{12}}$ **k** $\frac{\sqrt{6}\times\sqrt{24}}{\sqrt{27}\times 2\sqrt{3}}$ **l** $\frac{4\sqrt{90}\times 7\sqrt{8}}{5\sqrt{32}\times 6\sqrt{10}}$

20 Expand and simplify each expression. 4.15

a $-3\sqrt{2}\left(2\sqrt{2}-3\right)$ **b** $\sqrt{10}\left(1-5\sqrt{2}\right)$

c $\left(3\sqrt{5}-2\sqrt{7}\right)\left(3\sqrt{7}+\sqrt{5}\right)$ **d** $\left(\sqrt{7}-4\right)^2$

e $\left(5\sqrt{3}+\sqrt{2}\right)^2$ **f** $\left(3\sqrt{7}-2\sqrt{5}\right)\left(3\sqrt{7}+2\sqrt{5}\right)$

g $\left(3+4\sqrt{7}\right)\left(4\sqrt{7}-3\right)$ **h** $\left(5\sqrt{10}-3\sqrt{2}\right)\left(2\sqrt{10}+5\sqrt{2}\right)$

21 Rationalise the denominator of each surd. 4.16

a $\frac{1}{\sqrt{10}}$ **b** $\frac{3}{\sqrt{2}}$ **c** $\frac{1}{5\sqrt{7}}$

d $\frac{3}{4\sqrt{3}}$ **e** $\frac{5\sqrt{3}}{3\sqrt{2}}$ **f** $\frac{4+\sqrt{2}}{3\sqrt{2}}$

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